

NEUR 411 · NEUROETHOLOGY · FALL 2026

Lecture 12

Insect vision II: jumping spider eyes and predatory tracking (Land)

Monday, September 21, 2026

Jordan Lopez, Ph.D.

Professor of Neurobiology & Biomedical Engineering



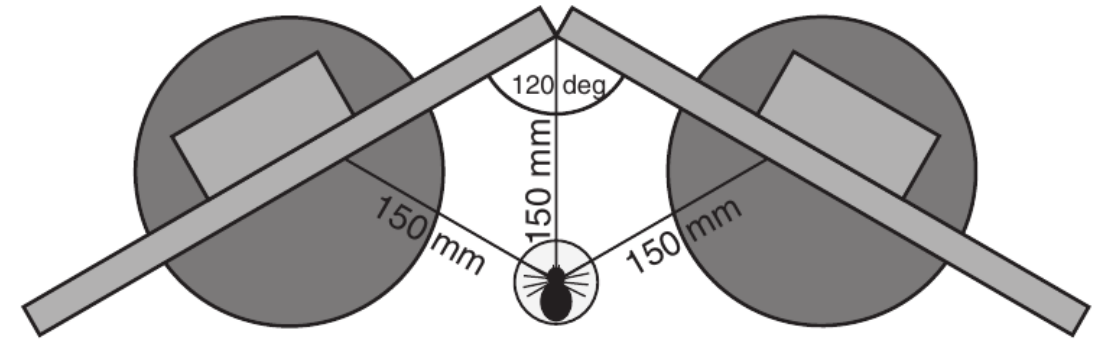
Zurek et al. (2015), Fig. 1(A). *Habronattus pyrrithrix*, a jumping spider studied for retinal spectral filtering.

Lateral eyes can guide stalking and attacks

Jumping spiders are salticid arachnids that approach prey before launching a directed jump. **Stalking** is the controlled approach preceding attack; **orientation** is a body turn toward a visual target. These actions can depend on different eye pairs.

Zurek and colleagues covered every eye except the **anterior lateral eyes**, the frontmost lateral pair, in *Servaea vestita*. Spiders encountered a live fly on a sloping ramp, initially approximately 25 cm away. The manipulation isolated visual information available through that pair.

Spiders retaining only anterior lateral vision could orient, stalk and attack. Performance was limited, and some individuals did not pursue the fly. The result establishes that these eyes can support parts of predation; it does not establish that the principal eyes are dispensable for normal hunting.



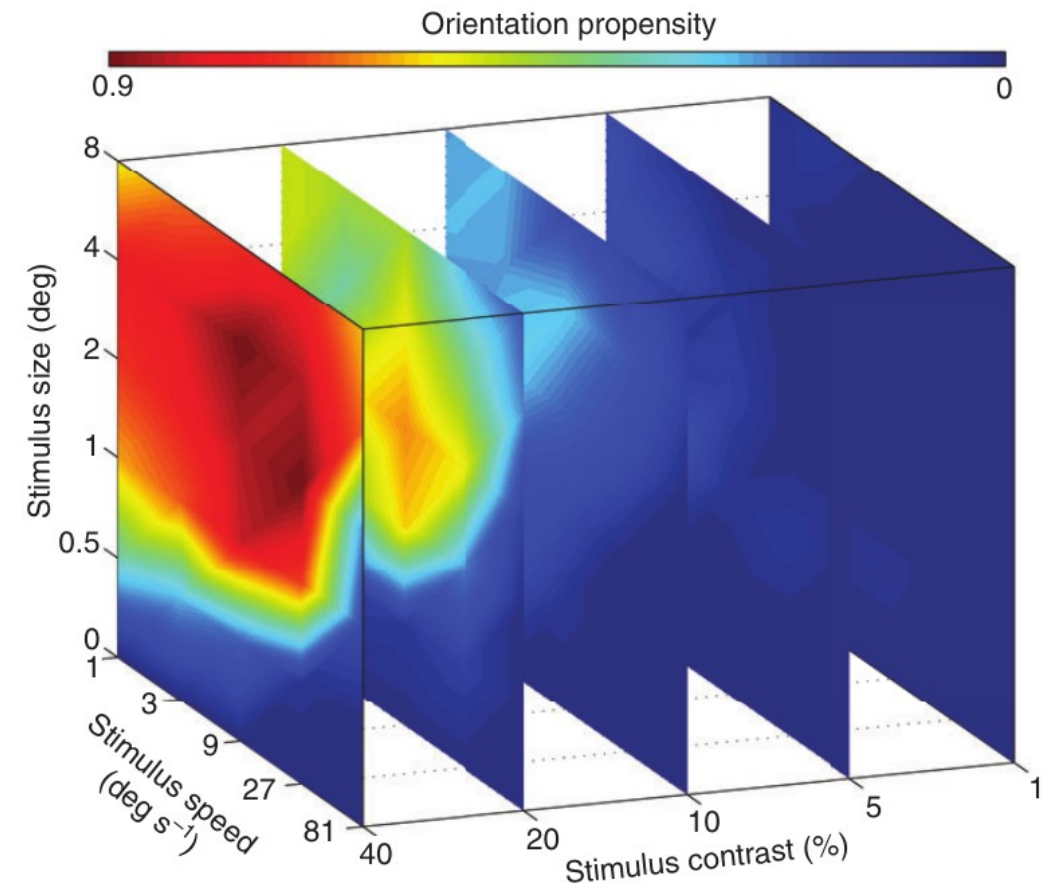
Zurek et al. (2010), Fig. 1. Controlled display arrangement for isolated lateral vision.

Controlled dots isolate visually evoked orientation

A **visual field** is the region of space accessible to an eye. To measure target detection within the anterior lateral field, Zurek and colleagues positioned *Servaea vestita* on a freely rotating 15-mm polystyrene ball while the spider remained fixed between two display screens.

The screens were 150 mm away and separated by 120°. Dark dots varied in angular diameter, speed and contrast against a gray background. **Angular size** specifies an object's extent at the eye in degrees, independently of its physical size.

A 4° dot moving at 9°/s with 40% contrast elicited orientation frequently, whereas a blank gray screen rarely did. The ball registered attempted turns without moving the spider through the scene. These responses measure detection and orientation, not recognition of a particular prey species.



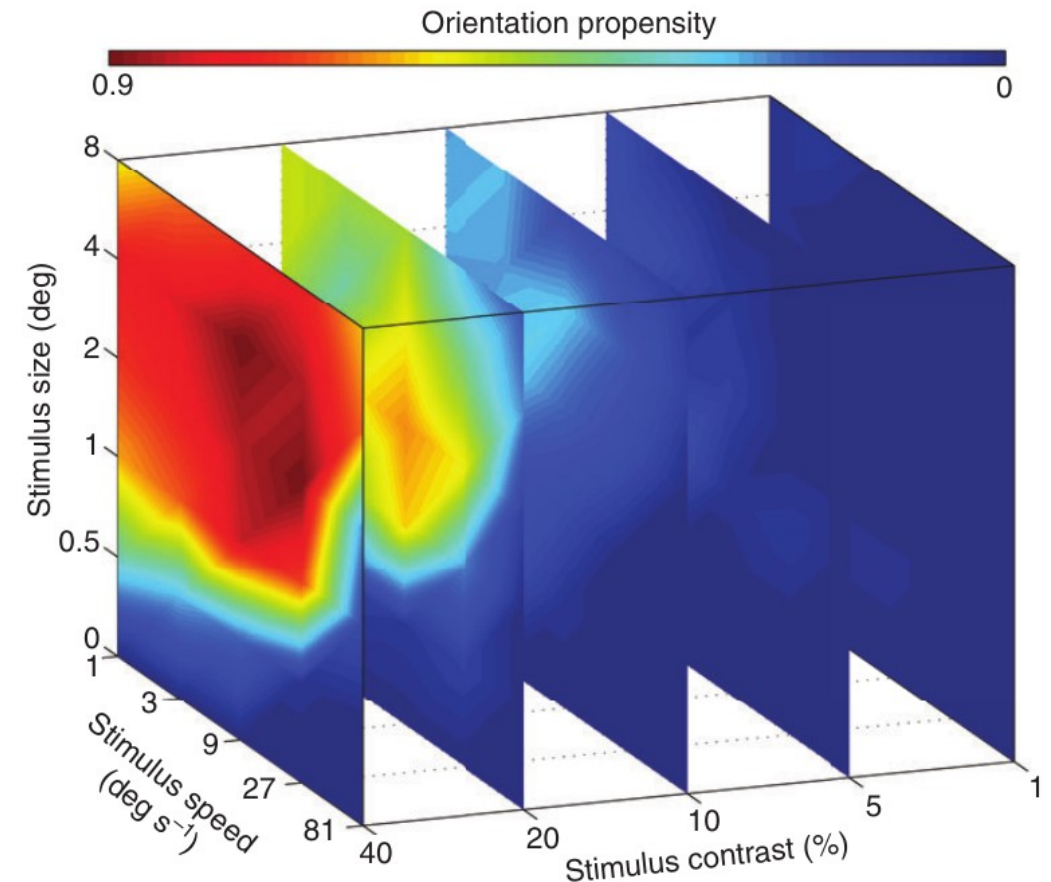
Zurek et al. (2010), Fig. 2. Orientation probability across size, speed and contrast.

Orientation depends jointly on size and speed

Contrast is the luminance difference between a target and its background relative to the background luminance in this experiment. Zurek and colleagues varied contrast alongside dot diameter, rather than treating every visible object as an equally effective stimulus.

Dots ranged from 0.5° to 8° , with speeds spanning 1 – $81^\circ/\text{s}$. Orientation depended on the combination of these variables: targets near 2 – 4° at stronger contrast were especially effective, but the favorable speed changed with target size and contrast.

Weak contrast reduced responses, and the smallest dot was not simply the best stimulus because it resembled distant prey. The measured response map constrains explanations of anterior lateral detection. It does not supply a single acuity threshold or a neural mechanism for multiplying size, contrast and motion signals.



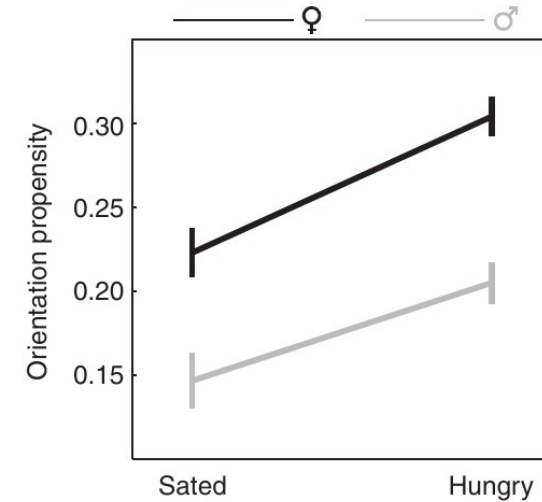
Zurek et al. (2010), Fig. 2. Size and speed interact at each tested contrast.

Hunger alters the probability of turning toward a target

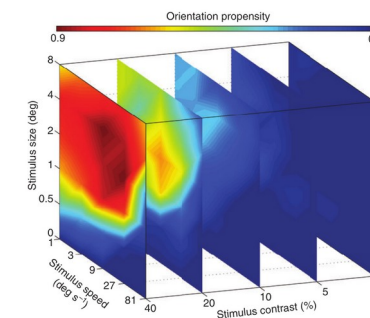
Internal state includes physiological conditions such as hunger that influence behavioral responses to the same sensory input. Zurek and colleagues compared *Servaea vestita* one day after feeding with spiders tested seven days after feeding.

Eye availability and moving-dot presentation were controlled while feeding history changed. Hungry spiders oriented more often, and females generally responded more frequently than males. The measured variable was the probability of a turn, not a direct receptor voltage or brain firing rate.

Hunger therefore changes the relationship between a visual target and an overt action. The experiment does not localize that change to photoreceptors, visual interneurons or motor circuits. Lower response probability in a recently fed spider cannot by itself be interpreted as failure to detect the target.



Zurek et al. (2010), Fig. 3. Feeding state and sex affect orientation.



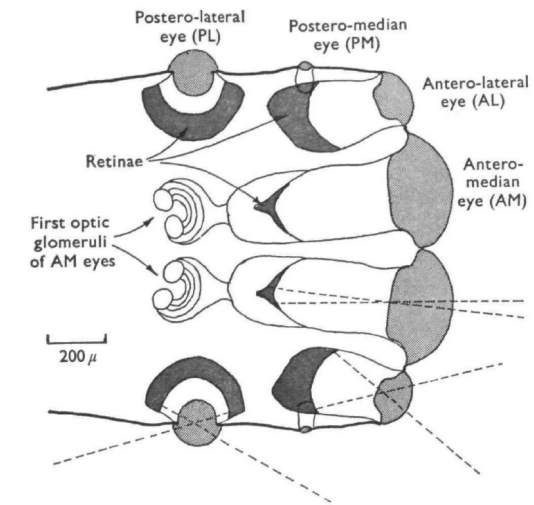
Zurek et al. (2010), Fig. 2. The controlled target conditions used in the behavioral assay.

Principal eyes magnify a restricted visual field

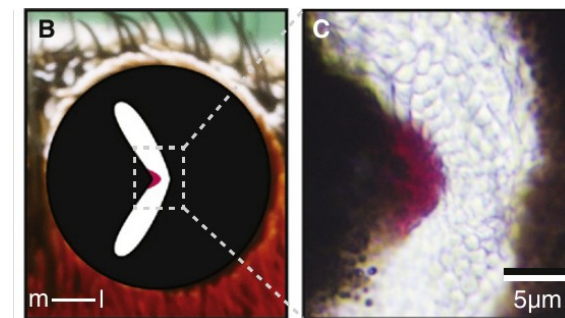
The **principal eyes**, also called anterior median eyes, are the large forward-facing pair. Their long optical tubes produce a magnified retinal image over a restricted field. Land compared their optical organization with the broader viewing arrangement of the anterior lateral eyes.

In *Phidippus* and *Metaphidippus*, the principal-eye lens remains fixed in the body while the retinal tube can move behind it. Moving that tube changes which direction is sampled by the small high-resolution receptor array.

A broad lateral field and a restricted principal field provide different spatial access to a target. The colored retinal preparation from *Habronattus* identifies a comparable principal-eye structure in another genus. Its red filter is a specialization of that lineage and should not be assigned to Land's species.



Land (1969a), Fig. 1. Principal and anterior lateral eye optics.



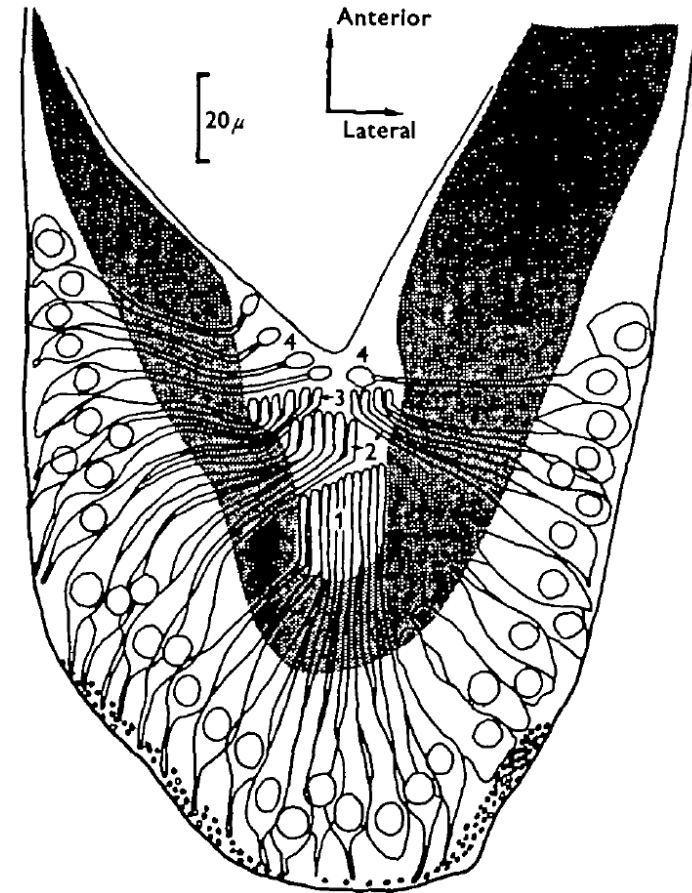
Zurek et al. (2015), Fig. 1(B, C). Principal-eye aperture and retinal filter in *Habronattus*.

Four retinal layers occupy different optical depths

A **photoreceptor** is a sensory cell whose light-sensitive region converts absorbed light into an electrical response. Land's principal-eye sections revealed four receptor layers at different distances from the lens, numbered from the deepest layer, L1, toward L4 nearer the incoming light.

L1 and L2 extend across the boomerang-shaped retina; L3 and L4 occupy more restricted central regions. Receptor segments in L1–L3 are largely aligned with incident light, whereas L4 contains differently oriented, more ovoid segments.

The retina is therefore a three-dimensional receptor arrangement rather than a single flat screen. Histological depth identifies possible differences in optical focus and light sampling. Anatomy alone does not establish a layer's wavelength sensitivity, response kinetics or contribution to a specific hunting decision.



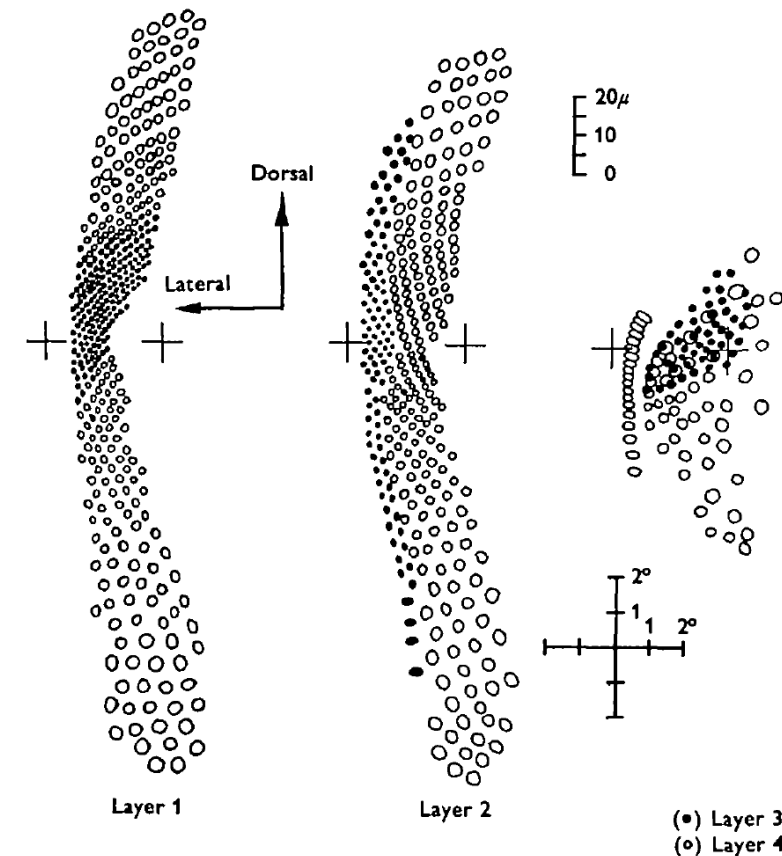
Land (1969a), Fig. 3. Layered principal-eye retina and its cellular organization.

Dense receptor spacing supports fine angular sampling

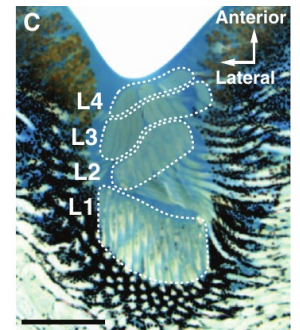
A **receptor mosaic** is the spatial arrangement of sensory cells sampling an image. Land measured especially dense packing in the principal-eye retinal region aligned with the highest-resolution viewing direction.

Minimum receptor separation was approximately $1.7\text{ }\mu\text{m}$, corresponding to about 11 arcmin in the studied eyes. An **arcminute** is one sixtieth of a degree. These measurements link microscopic spacing to the angular separation of neighboring image samples.

Fine sampling permits small spatial differences to reach separate receptor positions, but receptor spacing alone does not prove behavioral discrimination at that limit. Optical blur, neural pooling and viewing conditions can change effective acuity. The colored section identifies the corresponding layered organ in *Hasarius*, without implying that its spacing equals Land's measured value.



Land (1969a), Fig. 5. Receptor mosaics in the principal-eye layers.



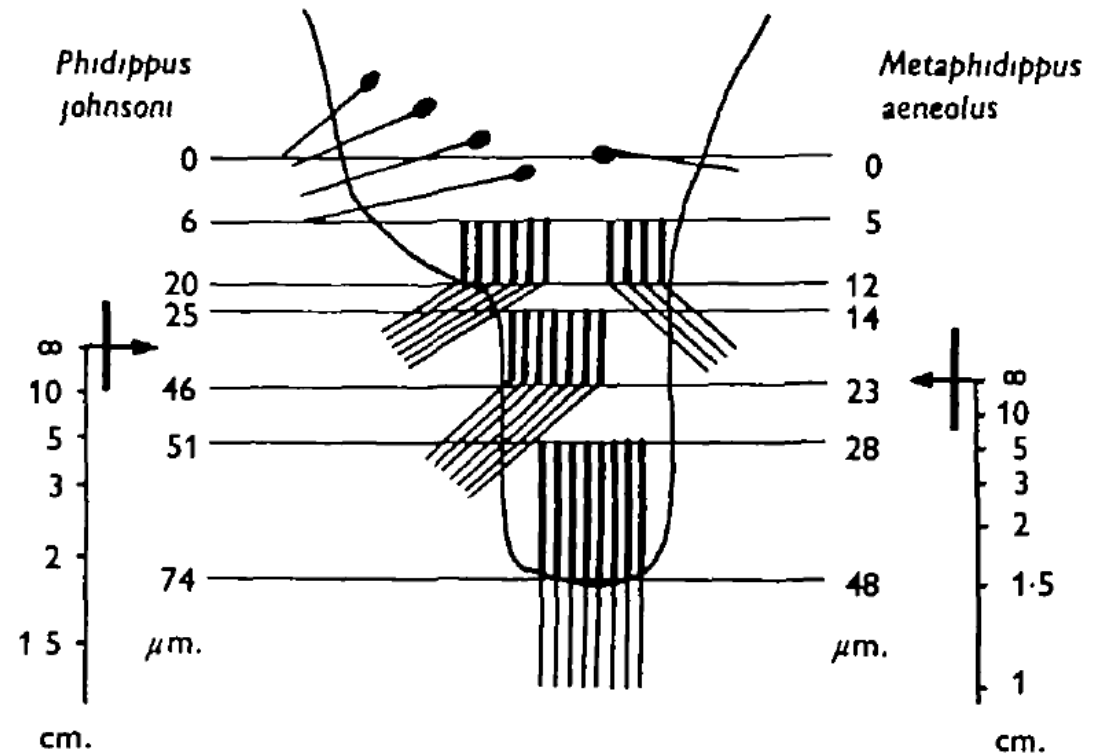
Nagata et al. (2012), Fig. 1(C). Four retinal layers in *Hasarius adansoni*.

Layer position changes where an image is in focus

Focus is the optical condition in which rays from an object form a comparatively sharp image at a receptor plane. A layered retina places different receptor populations at different image distances, so an object need not be equally sharp in every layer.

Land combined retinal anatomy with optical measurements. For blue-green light, the best image approached L2, while deeper L1 could receive a focused image from an object at approximately 2 cm under the reported optical conditions.

These relationships make retinal depth relevant to viewing distance. They do not prove that the spider actively accommodates its lens or consciously selects a layer. Land's proposed spectral assignments to the layers were hypotheses; later measurements of receptor sensitivity and pigment localization must determine which wavelengths actually activate each population.



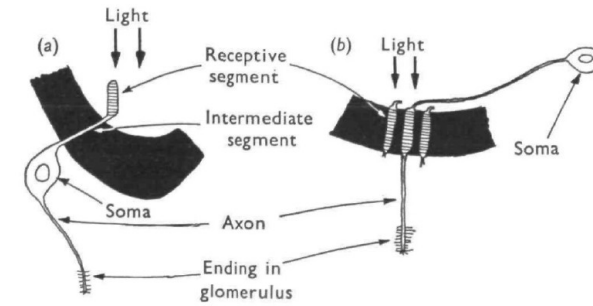
Land (1969a), Fig. 12. Image positions for different object distances.

Receptor projections preserve layered organization

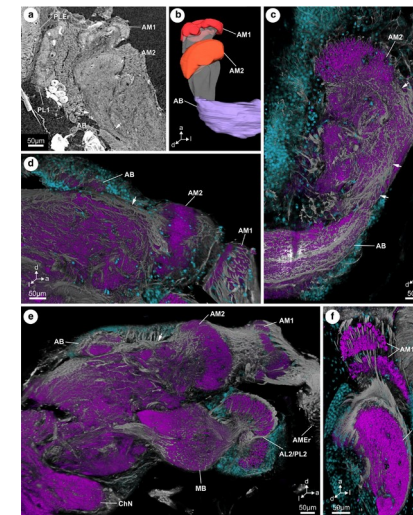
An **axon** is the neuronal process carrying signals away from a cell's sensory region. A **neuropil** is a dense region of neuronal processes where synaptic interactions commonly occur. Land traced principal-eye receptor fibers toward a first visual neuropil, historically termed the optic glomerulus.

Fibers arising from different retinal layers appeared to end in different regions of that neuropil. The result supports organized input rather than complete mixing immediately after light detection. It is based on anatomical tracing, not recordings of synaptic transmission.

Modern labeled sections in *Marpissa muscosa* identify AM1 and AM2, the first- and second-order principal-eye neuropils. They provide anatomical landmarks for a comparable pathway, while leaving the exact correspondence between Land's layer-specific endings and modern neuronal cell types unresolved.



Land (1969a), Fig. 2. Retinal layers and axonal endings in the first optic glomerulus.



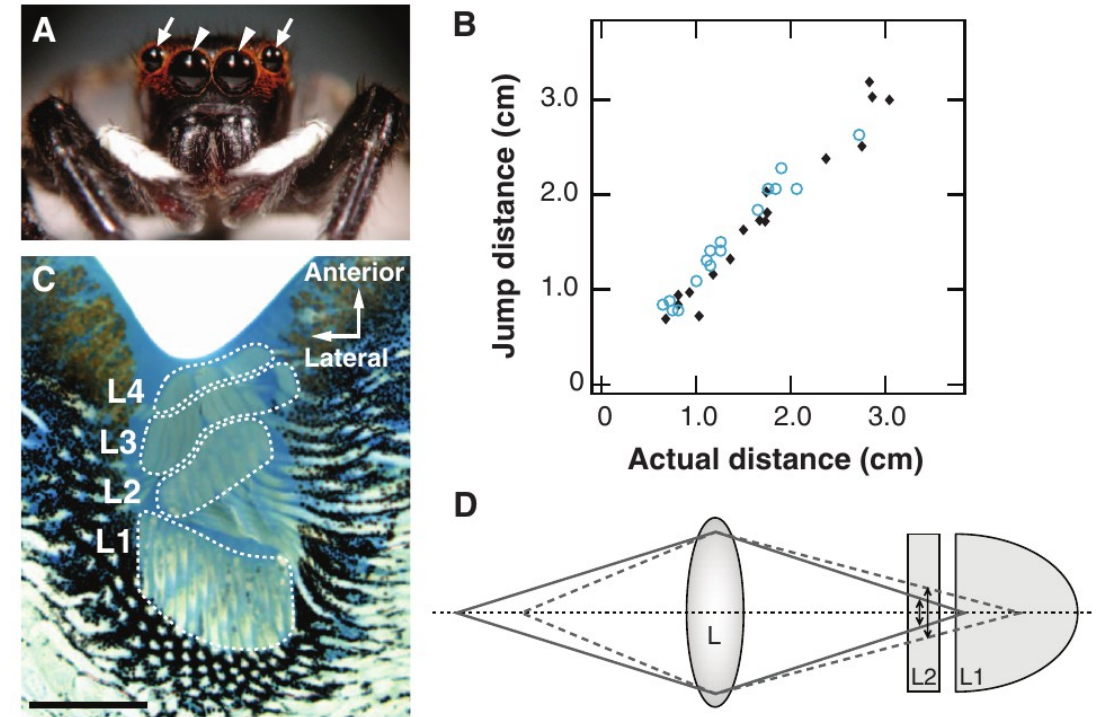
Steinhoff et al. (2020), Fig. 1(a–f). Principal-eye neuropils in *Marpissa muscosa*.

Wavelength changes the focal plane of the same eye

Chromatic aberration is a wavelength-dependent difference in optical focus. Nagata and colleagues investigated this property in the principal eyes of *Hasarius adansoni*, whose retina contains receptors at four depths behind the lens.

Green and red light form their sharpest images at different positions. A layer receiving a blurred green image can receive a different amount of blur from the same object under red illumination, even when the object has not moved.

The authors proposed that this wavelength-dependent **defocus**, the departure from a sharp image, could bias distance estimation. Retinal anatomy and optical calculations support the availability of such a cue. Establishing its behavioral use requires altering illumination and measuring the resulting jump, rather than inferring depth perception from a section alone.



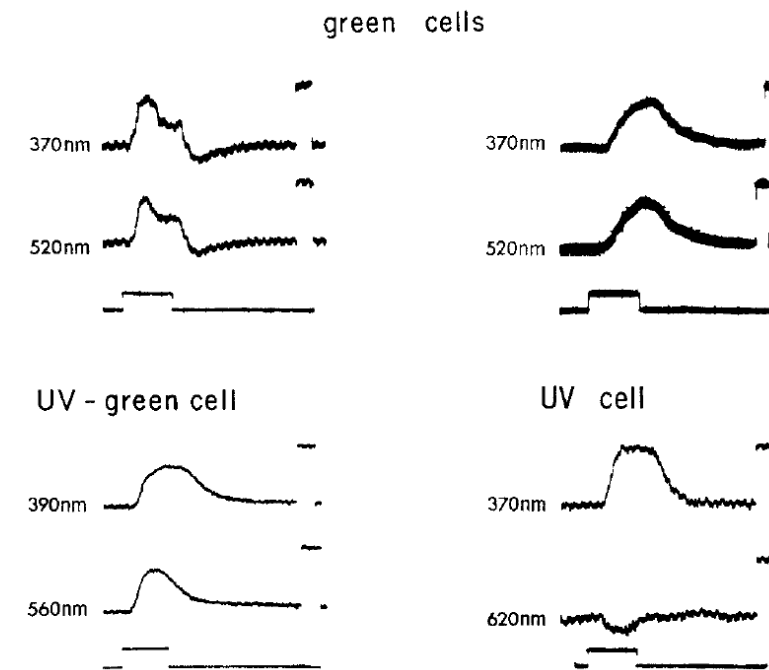
Nagata et al. (2012), Fig. 1(A–D). Principal-eye anatomy, jump accuracy and the published optical account.

Intracellular recordings resolve receptor voltage

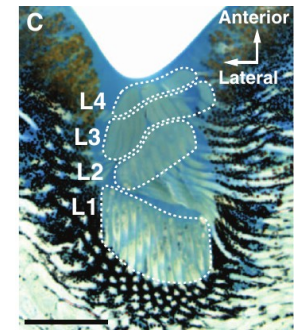
An **intracellular recording** measures voltage with an electrode inside a cell. DeVoe recorded principal-eye photoreceptors in *Phidippus regius* while presenting light flashes of selected wavelengths, linking light stimulation directly to receptor electrical activity.

The tested flashes lasted 100 ms. Responses were primarily graded depolarizations: the membrane voltage became less negative by an amount depending on the stimulus, rather than producing an all-or-none behavioral turn. The recording calibration included a 5-mV voltage reference.

Wavelength and intensity were varied to compare receptor sensitivity. The recordings did not identify the exact retinal layer of each impaled cell. The colored *Hasarius* section locates the same type of layered organ, but cannot retrospectively assign a *Phidippus* recording to one of its layers.



DeVoe (1975), Fig. 1. Responses to calibrated flashes in individual photoreceptors.



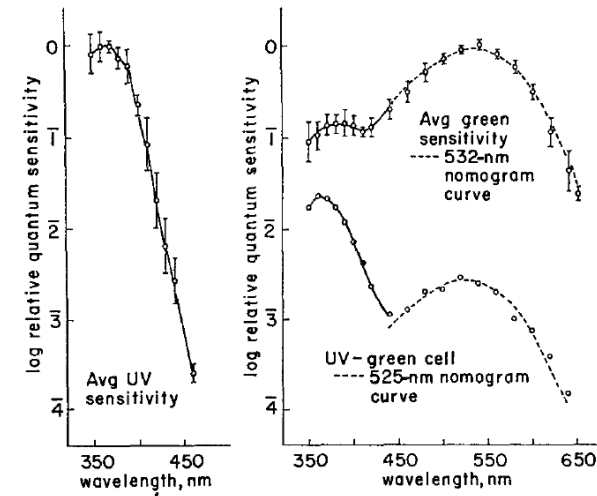
Nagata et al. (2012), Fig. 1(C). The layered principal-eye organ in *Hasarius*.

Ultraviolet receptors strongly favor short wavelengths

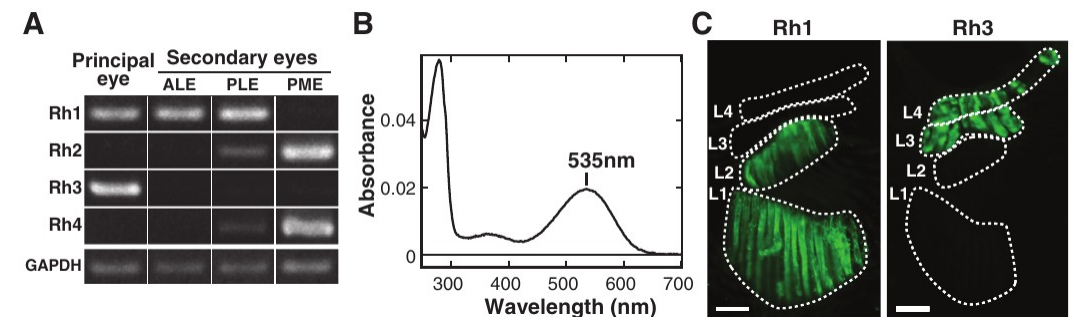
Spectral sensitivity specifies how effectively different wavelengths evoke a response after stimulus intensity is accounted for. DeVoe found principal-eye cells in *Phidippus regius* with maximum sensitivity near 370 nm, within the ultraviolet range.

These cells were more than four logarithmic units less sensitive at wavelengths beyond approximately 460 nm. A logarithmic sensitivity axis represents multiplicative changes, so the decline identifies strong spectral selectivity rather than a small difference in response amplitude.

The receptor class supplies a physiological channel for ultraviolet information. It does not establish ultraviolet color discrimination by the intact spider. Later pigment localization in *Hasarius* places an ultraviolet-sensitive pigment in upper retinal layers, but that molecular result cannot identify the unlocalized cells in DeVoe's earlier recordings.



DeVoe (1975), Fig. 2. Ultraviolet, green and dual-peaked spectral sensitivity.



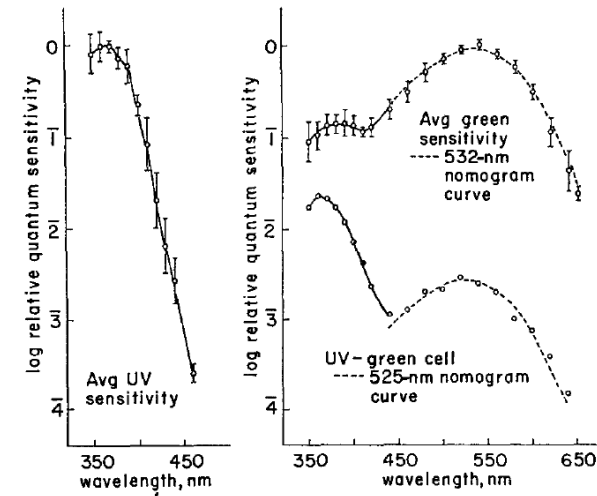
Nagata et al. (2012), Fig. 2(A–C). Pigment spectra and layer localization in *Hasarius*.

Green receptors have a distinct sensitivity peak

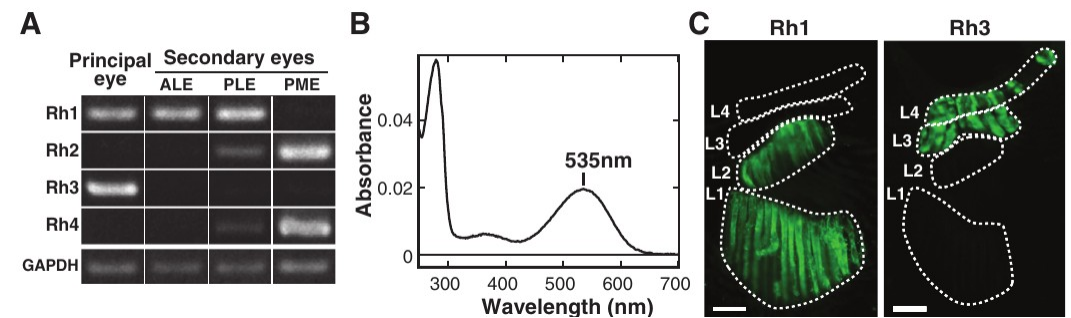
DeVoe also recorded a green-sensitive principal-eye receptor class in *Phidippus regius*. Its spectral sensitivity peaked near 532 nm and followed the expected shape of a visual-pigment sensitivity curve over much of the measured range.

A **visual pigment** is a light-absorbing molecular complex containing an opsin protein. An **opsin** is the protein component that contributes to wavelength selectivity. Nagata and colleagues later purified the *Hasarius* pigment Rh1 and measured an absorption maximum near 535 nm.

Receptor voltage sensitivity and purified-pigment absorption are complementary measurements. Their similar green peaks support a pigment-based contribution to green responses. The experiments involve different species and do not identify every molecular step between photon absorption, membrane current and the recorded voltage change.



DeVoe (1975), Fig. 2. Green sensitivity peaks near 532 nm.



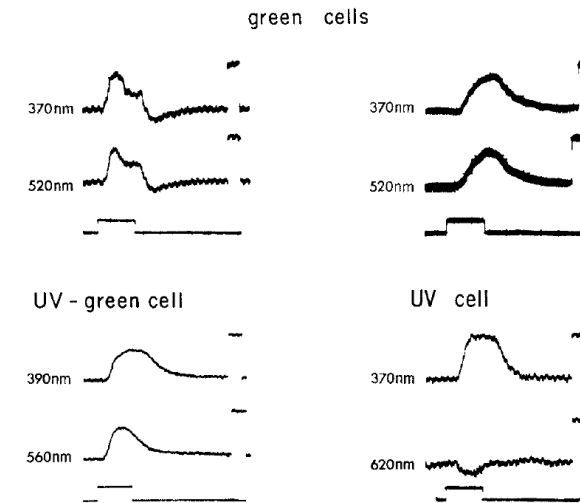
Nagata et al. (2012), Fig. 2(A–C). Rh1 pigment absorption and principal-eye localization.

Dual-peaked cells differ in response timing

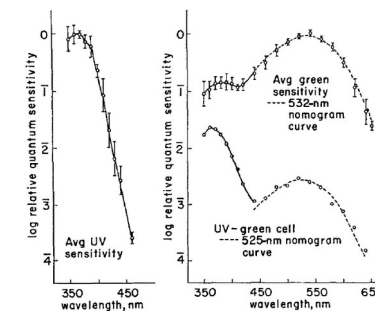
Some *Phidippus regius* cells responded strongly to both ultraviolet and green stimulation. Their spectral sensitivity had peaks near 370 nm and 525 nm, with relative ultraviolet-to-green sensitivity varying widely among the recorded cells.

In these **dual-peaked receptors**, ultraviolet responses were slower than green responses elicited from the same cell. The comparison held cell identity constant while changing wavelength, separating a difference within one receptor from a difference between two independently recorded populations.

The observed timing constrains accounts of signal generation in these cells. It does not establish two separately wired brain channels from that one receptor, or identify the molecular interaction responsible for the slower ultraviolet component. DeVoe also could not assign these cells to a particular retinal layer.



DeVoe (1975), Fig. 1. Responses of ultraviolet, green and ultraviolet-green cells.



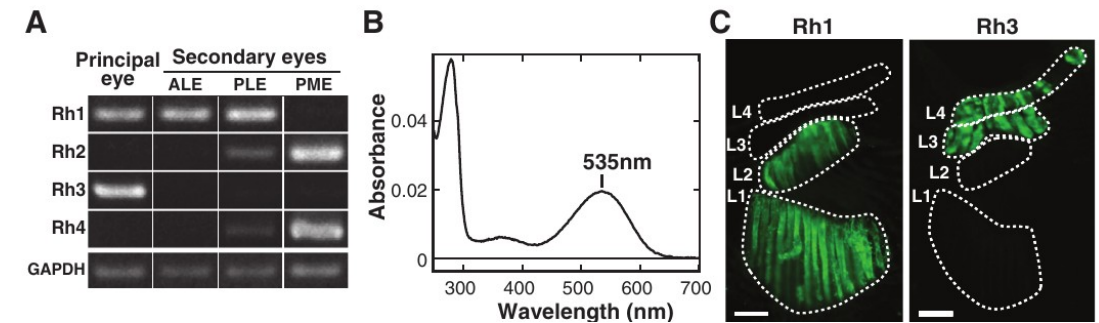
DeVoe (1975), Fig. 2. Dual-peaked sensitivity differs from the separate receptor classes.

Green pigment occupies both deeper retinal layers

Immunolabeling uses antibodies to identify the location of a molecular target in tissue. Nagata and colleagues combined principal-eye pigment expression measurements with antibody labeling in *Hasarius adansoni* to connect spectral identity with retinal depth.

Green-sensitive Rh1 was present through L1 and L2, while ultraviolet-sensitive Rh3 occurred in L3 and L4. The blue-sensitive pigment Rh2 was not detected in the principal-eye expression assay under the tested conditions. Rh1's purified absorption peak was approximately 535 nm.

L2 therefore contains green-sensitive receptors even though green images are not sharply focused there. This arrangement makes green-image blur biologically available. Absence from one expression assay does not prove a pigment is absent under every condition, and molecular localization does not by itself establish how downstream neurons use the layers.



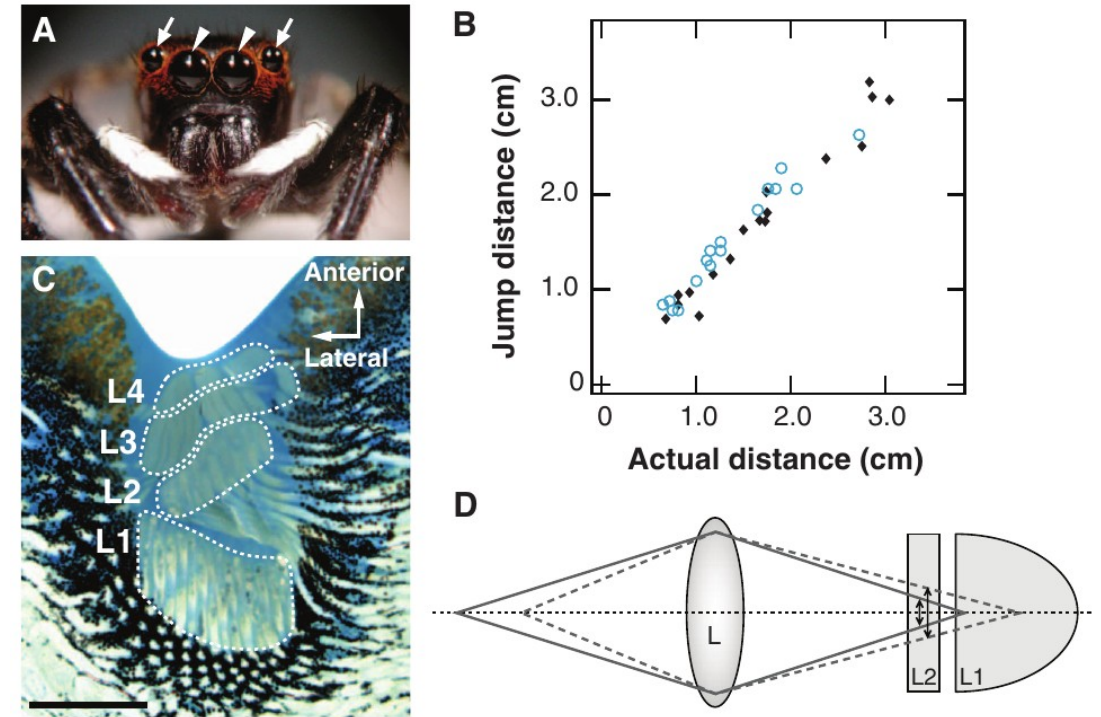
Nagata et al. (2012), Fig. 2(A–C). Rh1 in L1/L2 and ultraviolet-sensitive Rh3 in L3/L4.

One principal eye can support accurate prey jumps

Stereopsis estimates depth from differences between the views of two eyes. Nagata and colleagues tested whether two principal-eye views were necessary for accurate prey-directed jumps in *Hasarius adansoni*.

They covered both anterior lateral eyes and one principal eye, leaving a single principal eye available. Spiders still jumped accurately toward prey over the tested distances. Hunting observations also found no movement that would supply the proposed motion-parallax cue before these jumps.

Motion parallax is image displacement related to movement of the observer through space. Accurate performance with one principal eye weakens a requirement for binocular comparison in this preparation. The result does not, on its own, identify defocus as the remaining cue or exclude every possible monocular distance signal.



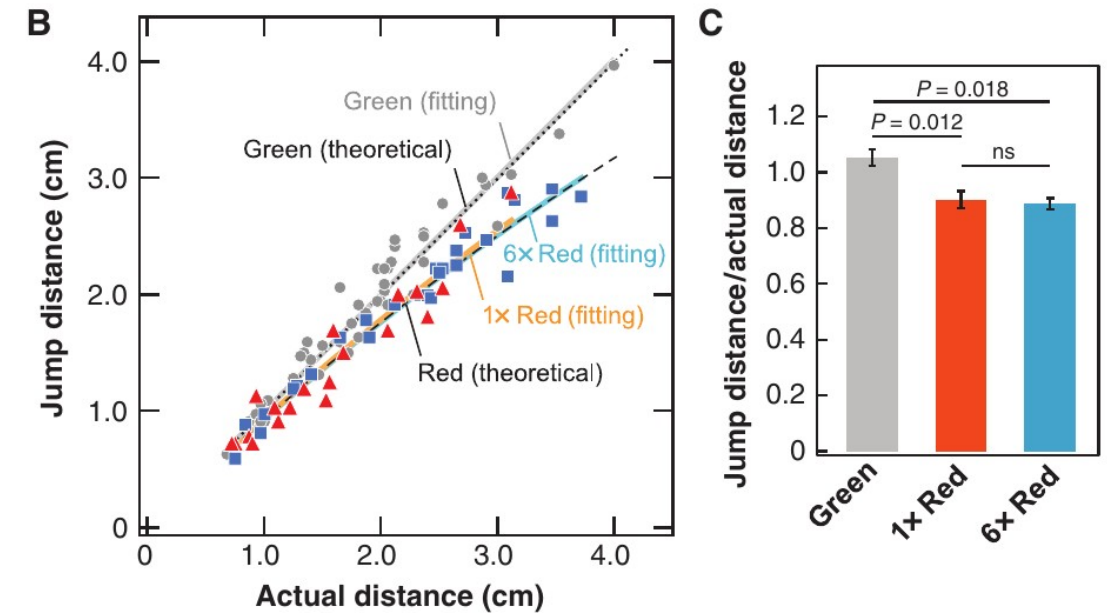
Nagata et al. (2012), Fig. 1(A–D). Eye occlusion and measured jump distance.

Red illumination makes prey jumps too short

Nagata and colleagues compared *Hasarius adansoni* jumping under approximately 520-nm green light and 630-nm red light. Wavelength changed the eye's optical focus while the experimenters measured the actual spider-to-prey distance and the ensuing jump.

Under green illumination, the average jump-to-target-distance ratio was approximately 1.0. Under the initial red condition it was approximately 0.90: spiders usually jumped short of the target. The recorded distances extended across the centimeter-scale hunting range.

A wavelength-dependent motor error supports a visual distance estimate affected by chromatic defocus. The result does not establish a red-sensitive receptor class or a preference for red prey. Green-sensitive Rh1 can absorb red light less efficiently, while the red light forms its image at a different optical depth.



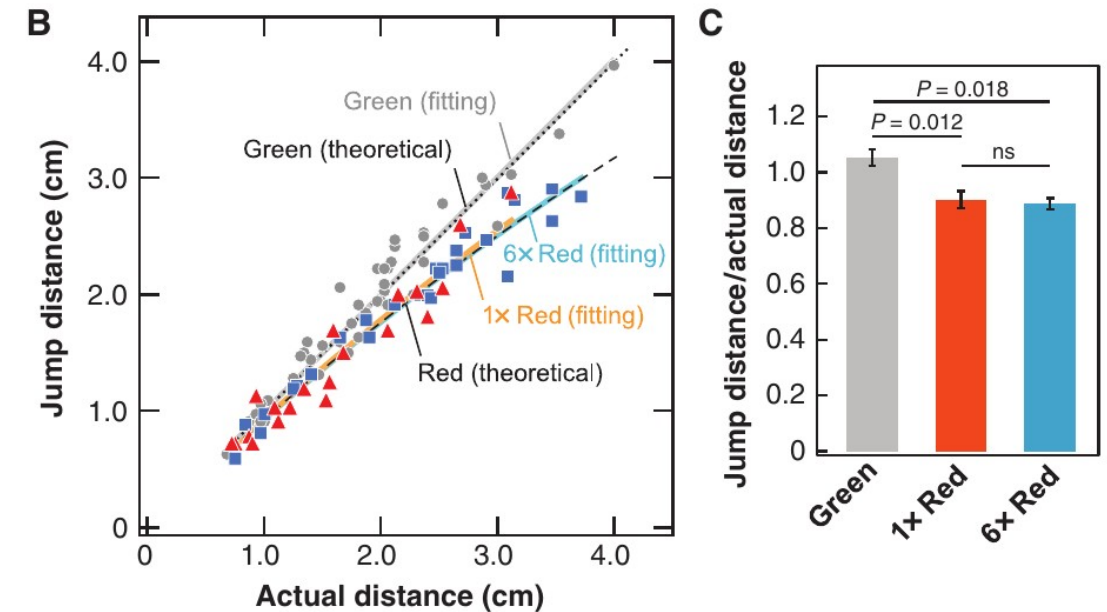
Nagata et al. (2012), Fig. 3(B, C). Measured jump distances under green and red light.

A brighter red control preserves the distance error

Changing wavelength also changes the effectiveness of light at activating a visual pigment. Nagata and colleagues adjusted the initial green and red conditions to stimulate purified Rh1 with comparable efficiency, then increased the red intensity sixfold as an additional control.

The jump-to-target-distance ratio remained approximately 0.89 under the brighter red condition, close to the approximately 0.90 ratio in the initial red condition. Greater red intensity therefore did not restore the green-light accuracy.

This control argues against reduced photon capture as a sufficient explanation of the short jumps. It supports a wavelength-dependent optical contribution. It does not identify the downstream computation, and intensity matching for purified Rh1 is not a complete measurement of every receptor's activity in the intact animal.



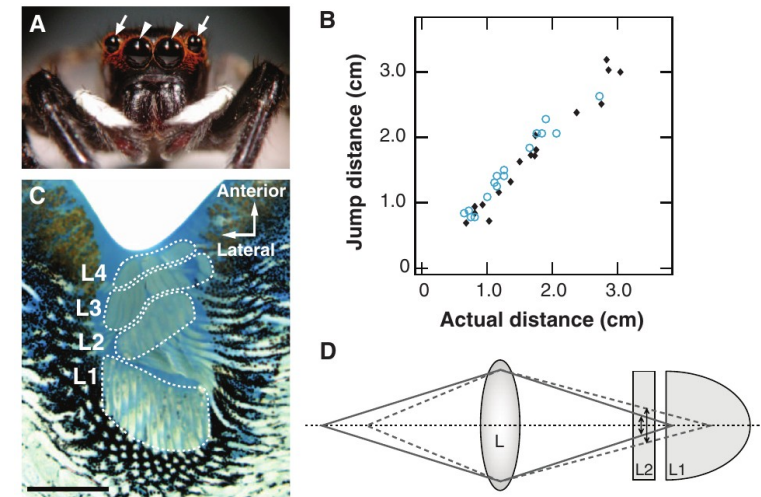
Nagata et al. (2012), Fig. 3(B, C). A sixfold red-light increase does not remove undershooting.

Defocused images provide a candidate distance signal

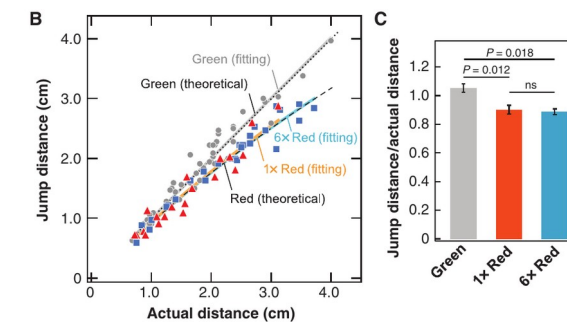
An image's blur changes with object distance when its receptor layer lies away from the sharp-image plane. Nagata and colleagues proposed that the amount of green-image defocus in L2 supplies an absolute distance cue in *Hasarius adansoni*.

Red light has a longer focal length in the measured optical system. Its additional blur would be interpreted as a closer target if the spider applies a green-calibrated relationship. The authors' optical predictions agreed with the direction of the observed red-light undershooting.

Comparing a sharper L1 image with a defocused L2 image was proposed as one possible implementation. The study explicitly lacked neural evidence for such a comparison. The anatomical layers, wavelength manipulation and jump measurements support the cue, while the comparing cells, synapses and ion channels remain unidentified.



Nagata et al. (2012), Fig. 1(A–D). Retinal depths and the authors' proposed defocus mechanism.



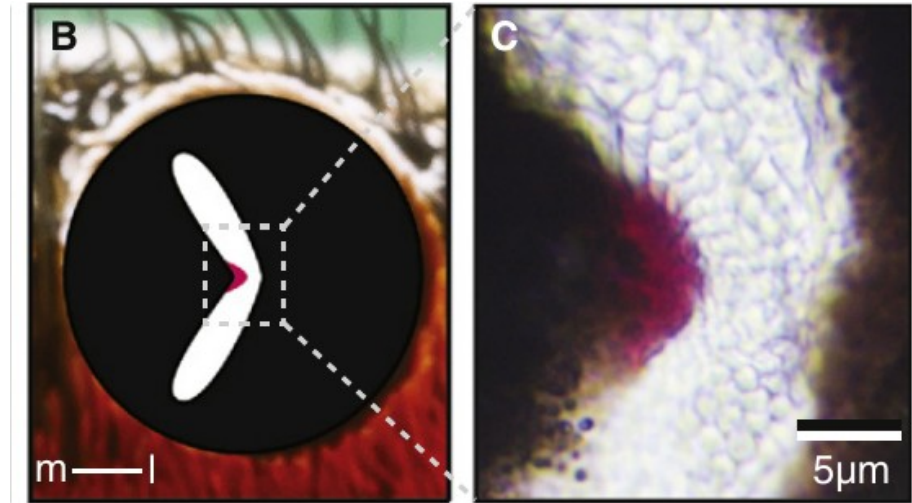
Nagata et al. (2012), Fig. 3(B, C). Published optical predictions alongside jump measurements.

A retinal filter shifts green sensitivity toward red

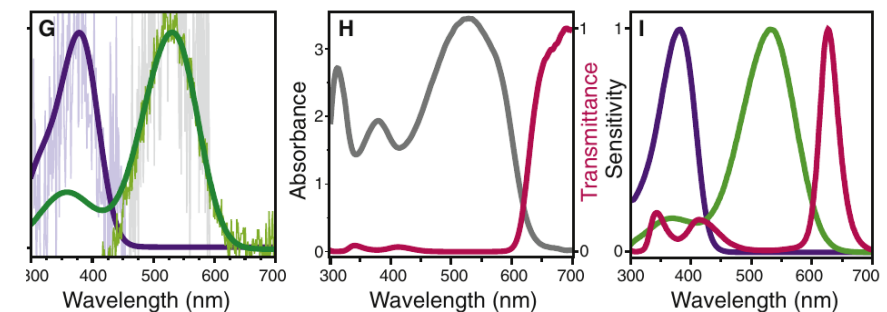
A **spectral filter** changes which wavelengths reach a receptor by absorbing some more strongly than others. Zurek and colleagues identified a red, long-pass filter in part of the principal-eye retina of *Habronattus pyrrithrix*. A long-pass filter preferentially transmits longer wavelengths.

The species has measured pigment peaks near 377 nm and 530 nm. Combining the green pigment's sensitivity with the measured filter transmission predicts a shifted sensitivity peak near 626 nm in filtered receptors, without requiring a third red-peaking opsin.

The filter supplies a physical route to an additional spectral channel. The predicted red curve is an author-published calculation from measured optical properties, not a directly recorded red-cell response. This specialization should not be assigned to every jumping-spider genus.



Zurek et al. (2015), Fig. 1(B, C). A localized red filter in the principal retina.



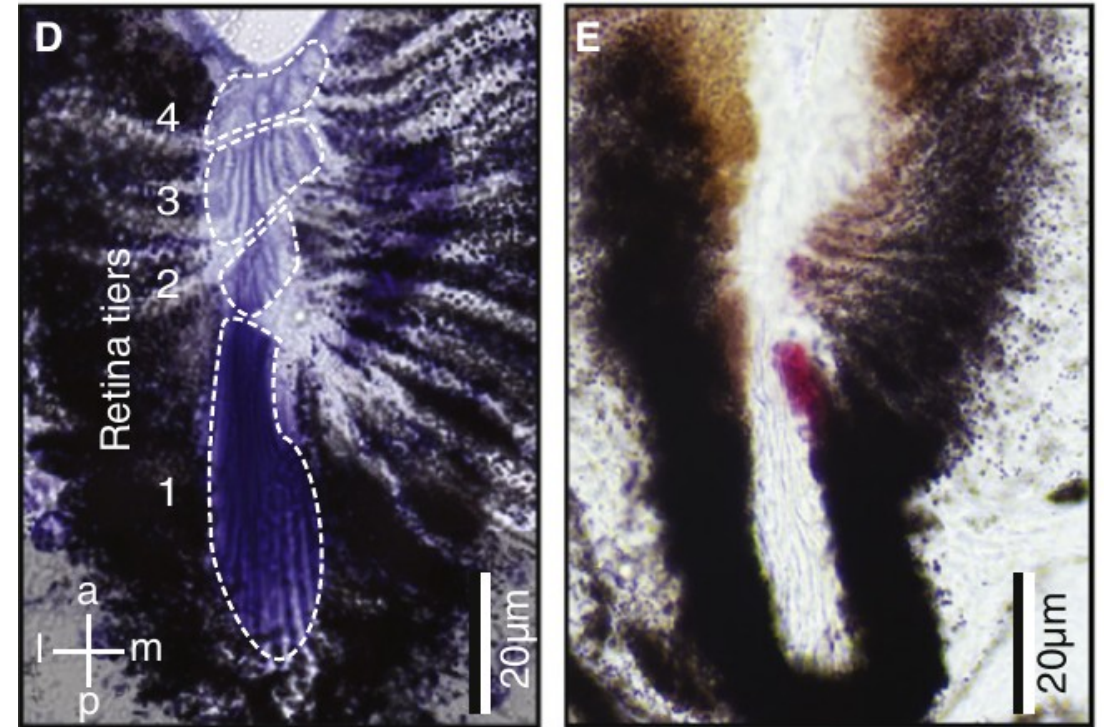
Zurek et al. (2015), Fig. 1(G-I). Measured spectra and predicted filtered sensitivity.

The filter covers a restricted high-acuity region

The red filter of *Habronattus pyrrithrix* is localized rather than spread across the entire principal retina. Unstained tissue sections retained the filter's native color, allowing its position to be compared with the receptor layers.

The colored region lies in a restricted central portion associated with L1 receptors. The published horizontal and coronal preparations used 14- μm sections; the layered retinal panels include 20- μm scale bars. This spatial arrangement permits filtered and unfiltered receptor populations within the same eye.

Filter localization makes spectral sampling depend on target alignment with a small retinal region. The authors found the feature in both sexes and other sampled *Habronattus* species, while sampled *Salticus* and *Phidippus* lacked it. These comparisons identify a lineage-associated specialization rather than a universal salticid design.



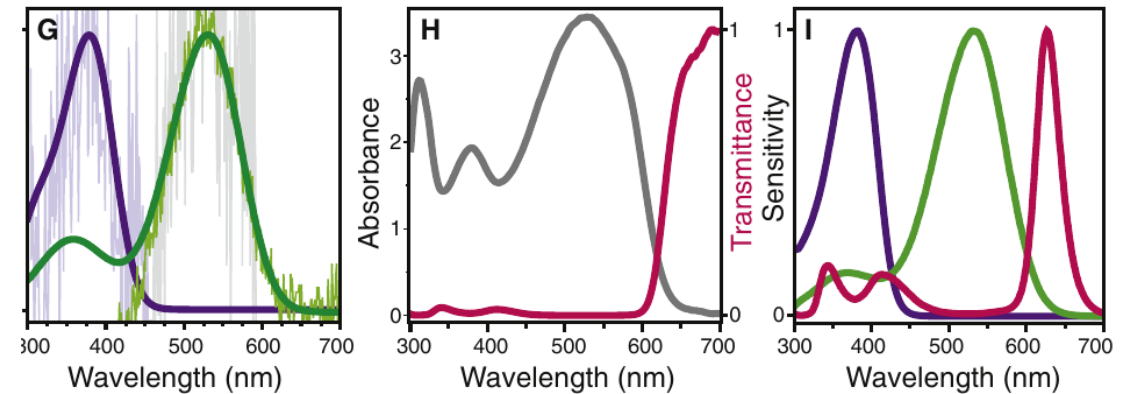
Zurek et al. (2015), Fig. 1(D, E). Layered retina and the restricted red-filter region.

Filtering trades photon capture for spectral separation

Trichromacy requires three functionally distinguishable spectral channels, rather than three differently colored structures. Zurek and colleagues proposed ultraviolet, unfiltered green and filtered red channels for *Habronattus pyrrithrix* from pigment and filter measurements.

The predicted green and red sensitivities separate because the filter removes much of the light otherwise absorbed by the green pigment. The same absorption that shifts sensitivity also reduces available photons, creating a potential cost under dim illumination.

The paper establishes the optical mechanism and its spectral potential. It does not directly measure three downstream opponent channels or demonstrate brightness-controlled discrimination among all relevant colors. **Spectral opponency**, comparison between wavelength channels, would require additional neural evidence before assigning a particular color computation to the brain.



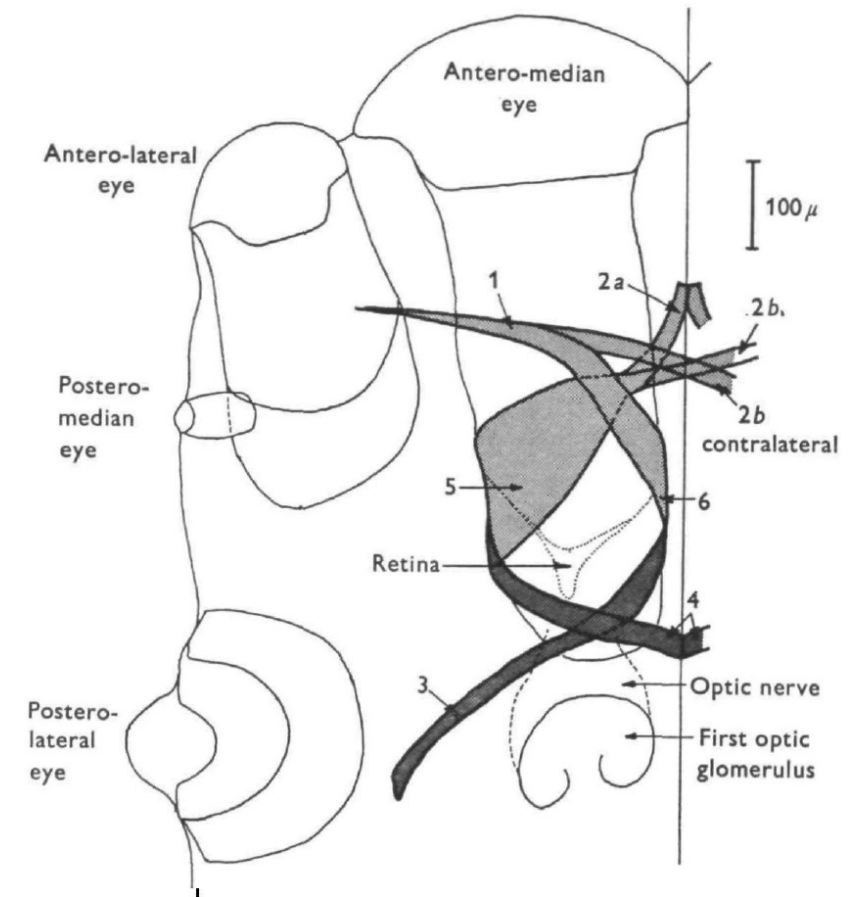
Zurek et al. (2015), Fig. 1(G–I). Filter absorbance, transmission and predicted receptor channels.

Six muscles reposition the principal-eye retina

The principal-eye retina is moved by muscles attached to its optical tube. Land's dissections in *Metaphidippus* identified six muscles associated with each principal eye, providing a mechanical basis for internal eye movements behind a fixed external lens.

Four muscles displace the retinal assembly in horizontal or vertical directions. Two thinner bands curve around the tube and can rotate it about its viewing axis. **Torsion** is this rotation around the optical axis, distinct from shifting the direction of gaze.

The muscle arrangement makes translation and rotation physically possible. Attachment geometry does not determine the precise activation sequence during tracking, and the dissections do not identify the transmitter or postsynaptic receptor at the neuromuscular junction. Behavioral movement records are needed to associate this anatomy with particular responses.



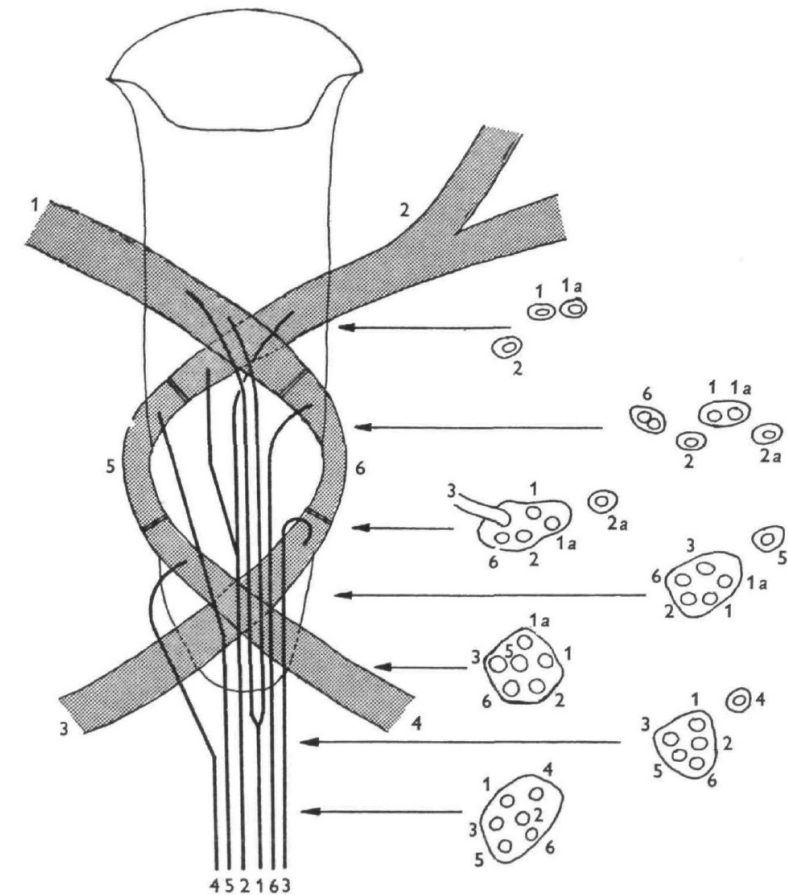
Land (1969b), Fig. 2. Six principal-eye muscles around the movable retinal tube.

Six motor axons connect to the eye muscles

A **motor axon** carries neuronal output to a muscle. Land traced an oculomotor nerve containing six axons toward the six principal-eye muscles, linking the movable retinal tube to a dedicated motor supply.

Histological tracing associated each axon with one muscle. The nerve and muscle arrangement separates the anatomical route for controlling retinal displacement from the photoreceptor fibers carrying sensory input toward the first optic glomerulus.

A one-to-one anatomical association between axons and muscles does not establish a six-command coding scheme for gaze. The study did not record these axons during scanning or identify their transmitter, receptor or membrane conductances. Sensory-triggered eye movements therefore require additional, unmeasured circuitry between visual input and this motor output.



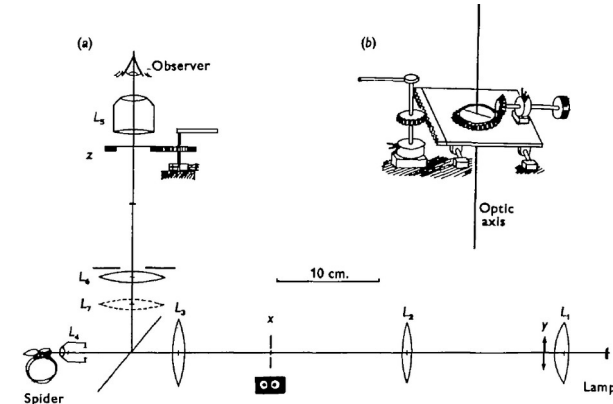
Land (1969b), Fig. 4. Oculomotor axons and their muscle terminations.

Ophthalmoscopy measures motion of the internal retina

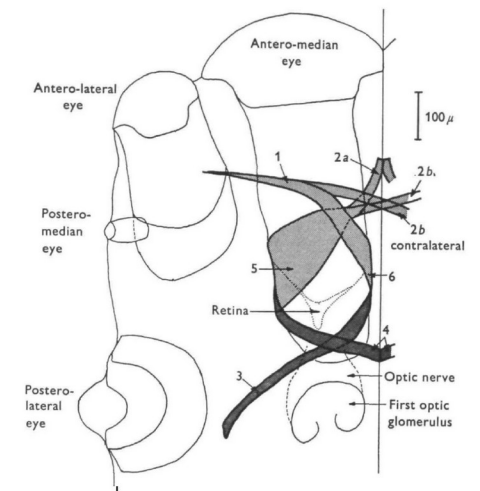
Ophthalmoscopy views structures inside an eye through its own optics. Land adapted this method to observe the principal-eye retina while a restrained spider viewed controlled targets, making internal motion measurable despite the fixed external lens.

The retinal field was followed relative to a reference, and position was recorded over time. Calibration related record displacement to angular movement; horizontal traces included a 10° reference and time marks, allowing movement amplitude and duration to be compared.

The method recorded retinal position, not receptor voltage or motor-neuron spikes. Restraint and the apparatus also differed from free hunting. Nevertheless, separate target and movement records distinguished spontaneous activity, rapid repositioning, tracking and scanning within the same anatomical organ.



Land (1969b), Fig. 1(a, b). Ophthalmoscopic recording apparatus.



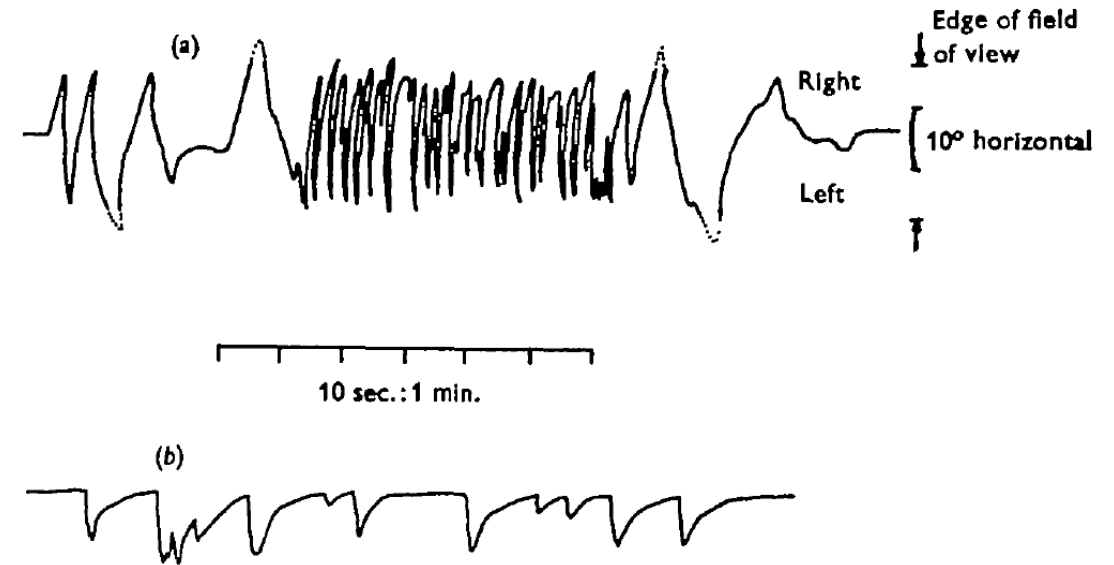
Land (1969b), Fig. 2. The movable retinal tube and muscle attachments.

Spontaneous shifts differ from visual saccades

A **saccade** is a rapid shift that repositions a visual sampling region. Land observed principal-eye movements both without an effective target and after small target displacements, distinguishing ongoing motor activity from visually evoked repositioning.

Spontaneous activity included irregular shifts and periods of rapid movement. It persisted when the visual field was blank and could occur after eyes were covered. Small stimulus movements instead evoked abrupt retinal shifts toward the target followed by a slower return.

Persistence without ordinary visual input supports an internally generated component of eye movement. The targeted responses establish visual influence on repositioning, but neither result identifies the central generator or the error signal. A spontaneous shift and a visually evoked saccade should not be treated as the same stimulus-dependent process.



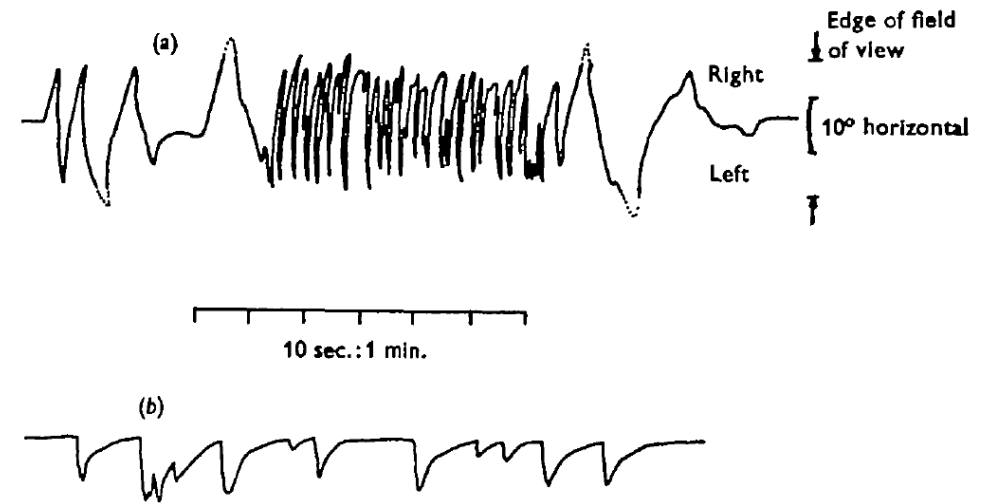
Land (1969b), Fig. 7(a, b). Spontaneous movements and stimulus-evoked saccades.

Moving targets can recruit retinal tracking

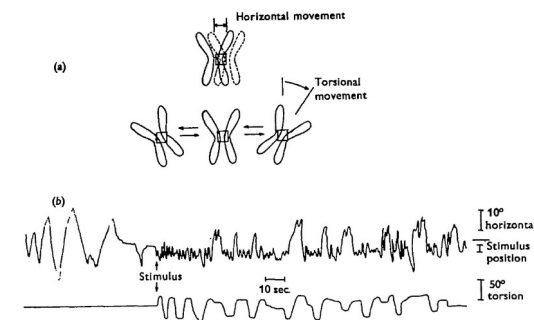
Tracking maintains a visual target near a sampled retinal region as its position changes. Land distinguished target-linked principal-eye tracking from spontaneous movement and the repeated scans performed over a relatively stationary object.

Target movements could evoke retinal repositioning when visible to the principal eyes or the anterior lateral eyes. This observation connects a broad-field sensory input to movement of the restricted principal-eye retina without requiring the entire spider to turn.

A visually guided change in retinal position establishes functional coordination between sensory input and the eye motor system. It does not identify the intervening neurons, prove a continuous velocity-matching controller or establish that principal and lateral inputs make identical contributions. Later eye masking can test the dependence on each input more directly.



Land (1969b), Fig. 7(a, b). Spontaneous and target-directed retinal shifts.



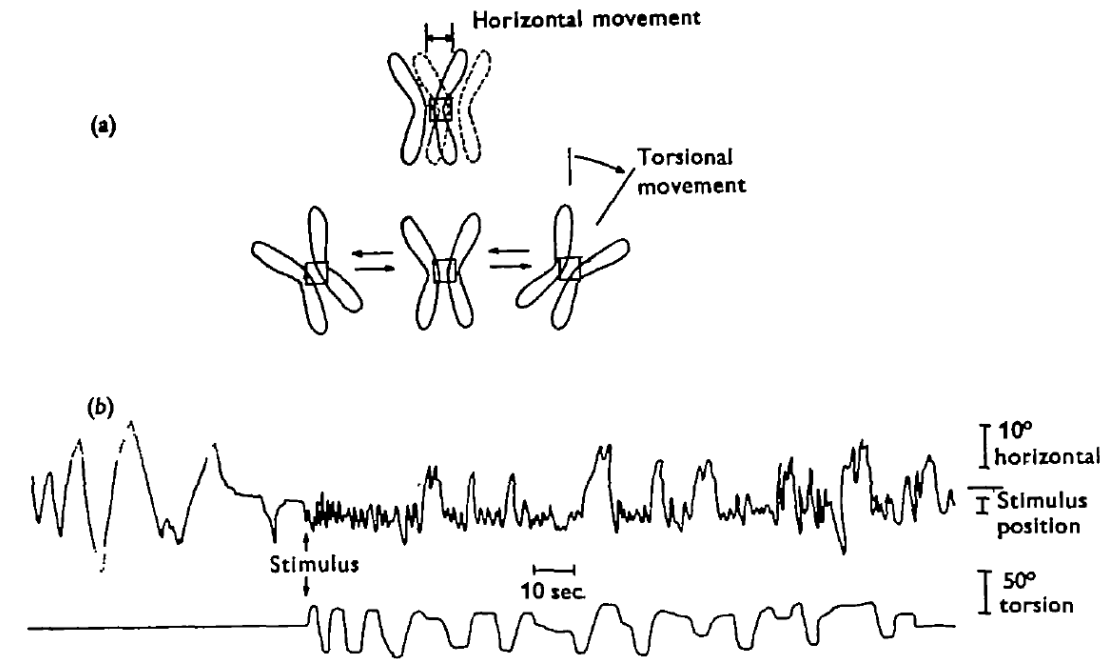
Land (1969b), Fig. 8(a, b). Horizontal and torsional components of eye movement.

Scanning combines horizontal movement with torsion

Scanning is repeated retinal movement over a target rather than a single shift toward it. Land recorded horizontal oscillations together with a slower rotation of the principal-eye retina about its optical axis during target examination.

Horizontal scans typically had periods of approximately 1–2 s. Torsional excursions extended through roughly 50° and developed over approximately 5–15 s. The distinct time scales allow receptor rows to sweep across a target while their orientation changes more slowly.

These movements alter the spatial samples delivered by a small retinal array. Their timing supports an active sampling strategy, but it does not establish what features a downstream neuron extracts. The electrical responses of individual receptors or feature-selective interneurons were not measured in these movement records.



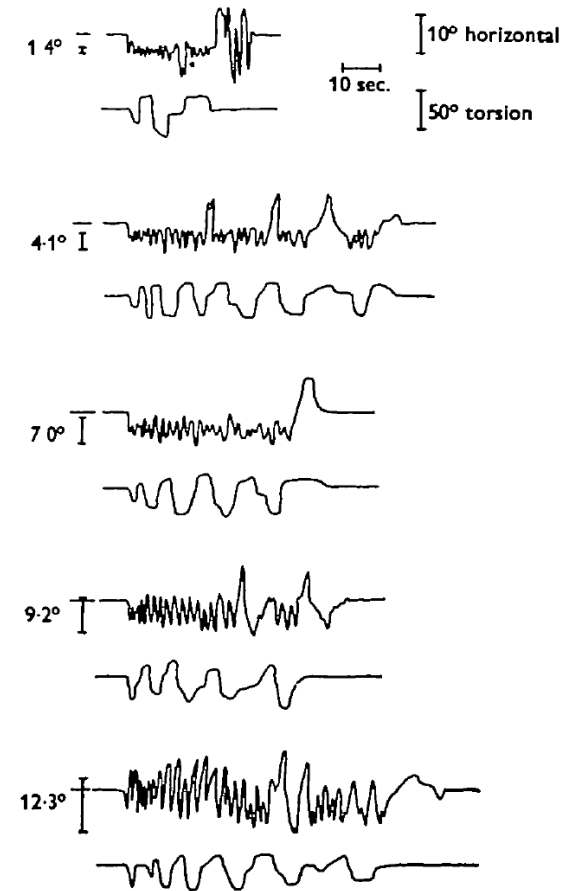
Land (1969b), Fig. 8(a, b). Published records separating scan translation and torsion.

Target width changes scan amplitude

Land varied target width while measuring principal-eye scanning. The published records include targets 1.4° , 4.1° , 7.0° , 9.2° and 12.3° wide, separating the size of the visual object from the retina's movement response.

For targets narrower than about 10° , horizontal scan amplitude was close to target width, with a minimum excursion near 2° even for a much smaller target. Wider targets were not always swept completely. Torsional excursion changed less systematically with width.

Land proposed that changing receptor-row orientation could aid contour analysis during these scans. That proposal drew on stimulus shapes known to alter behavior, but no line-selective neuron was recorded. Width-dependent scanning is measured evidence; feature extraction by a particular detector remains a mechanistic hypothesis.



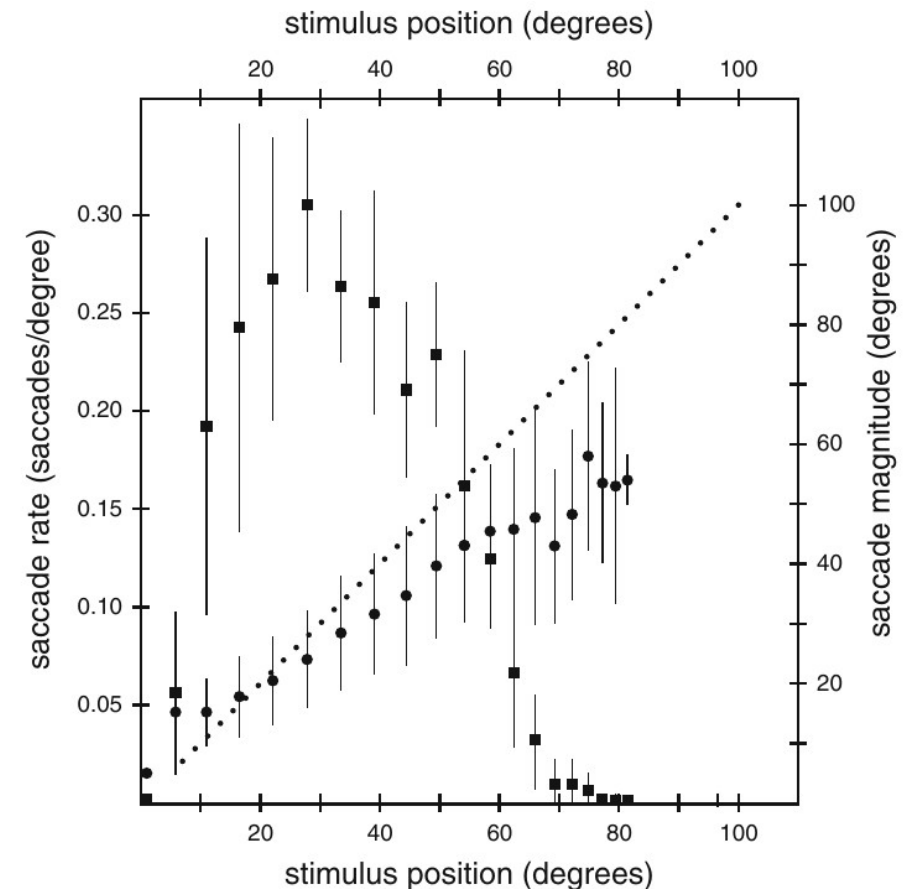
Land (1969b), Fig. 9. Scanning during targets of five angular widths.

Lateral eyes guide sequential body saccades

Zurek and Nelson isolated anterior lateral vision in *Servaea vestita* by covering the other eye pairs. A 4° dot moving at 9°/s elicited body turns measured while the spider remained fixed above a rotating ball.

Orientation consisted of a sequence of saccades rather than one smooth turn. The first response usually began after the entire dot entered the visible field. Near lateral target positions of 25–30°, the response rate reached approximately three saccades per second under this stimulus condition.

Body saccades redirect the animal toward a moving target and can place it nearer the principal-eye field. Their timing differs from Land's slower internal scans. The result establishes the anterior lateral contribution to a defined orienting task, not a complete model of free-moving prey interception.



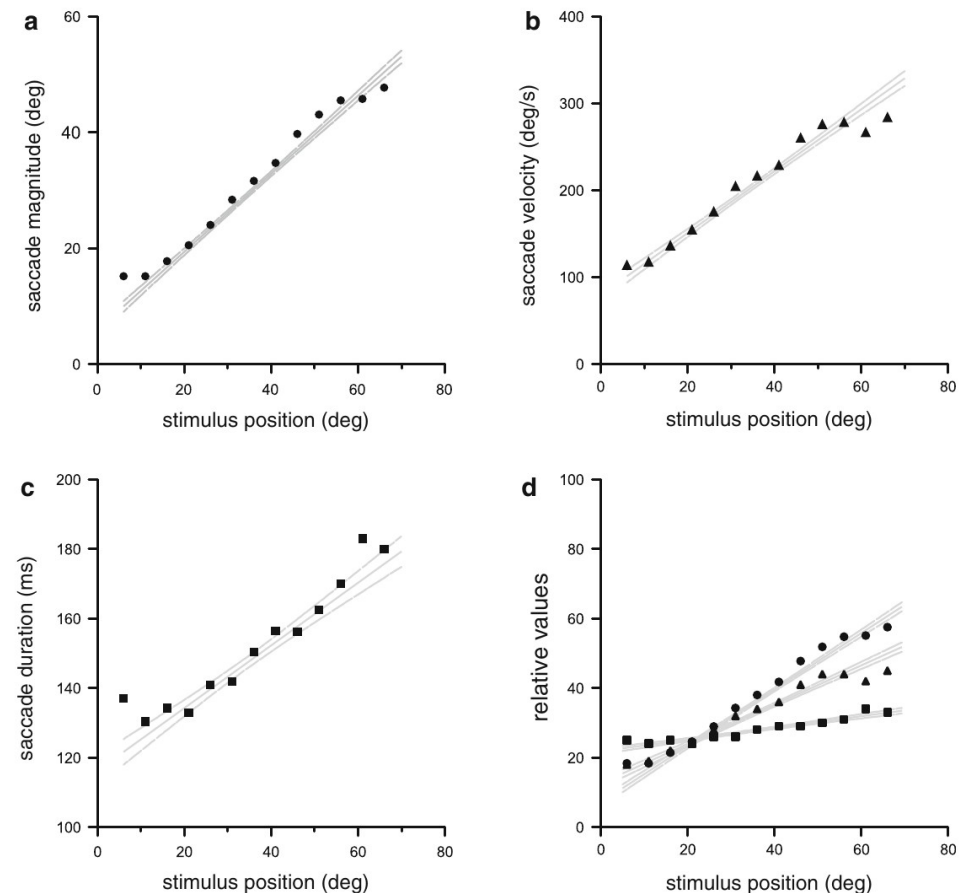
Zurek & Nelson (2012), Fig. 2. Saccade characteristics across horizontal target position.

Larger orienting saccades are executed faster

The **angular error** in an orienting task is the separation between the target direction and the animal's forward axis. Zurek and Nelson related body-saccade magnitude, velocity and duration to target position in lateral-eye-isolated *Servaea vestita*.

Targets farther to the side elicited larger and faster saccades. At positions beyond approximately 20°, the turns increasingly undershot the target direction. Higher speed partly limited the additional duration required for a larger movement, although it did not remove all duration differences.

A sequence of undershooting movements can progressively center a moving target. These relationships constrain motor control without proving a specific error-correcting neural circuit. The records are attempted body rotations on a ball and do not measure the forces or trajectories of an unrestricted spider.



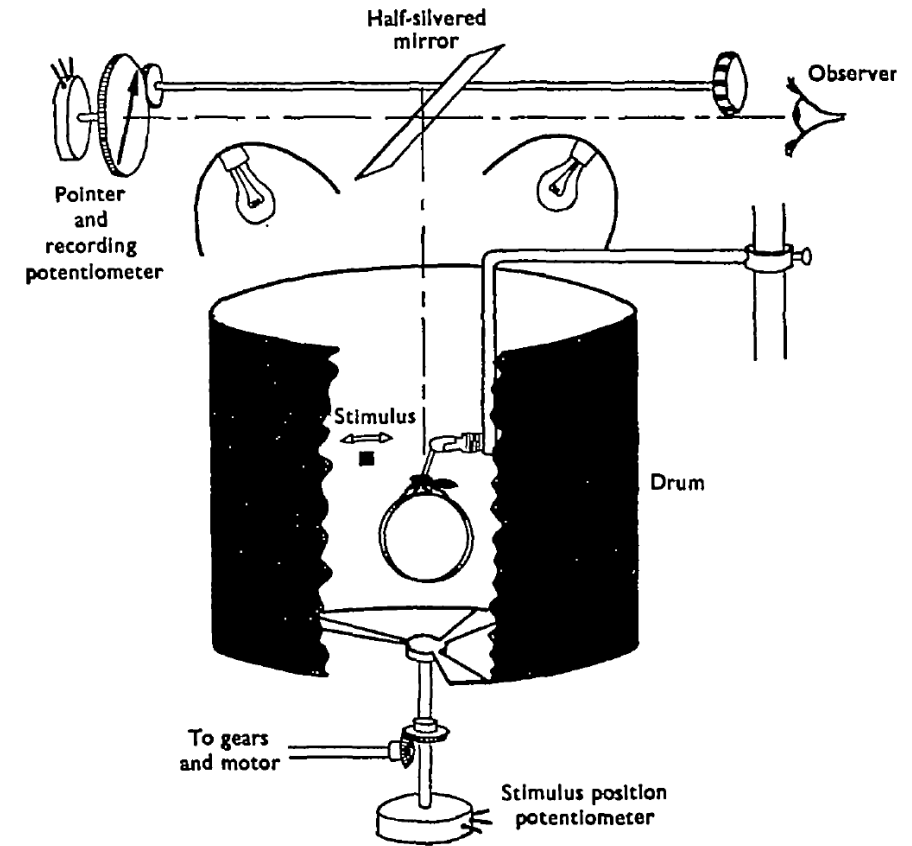
Zurek & Nelson (2012), Fig. 3(a-d). Magnitude, speed and duration vary with target position.

A rotating ring removes feedback from the body turn

Visual feedback is information about the consequences of an action returning through vision. Land tested orienting turns in *Metaphidippus* while holding the spider's body fixed and allowing its feet to rotate a lightweight ring.

A moving spot in the surrounding drum triggered attempted turning. The ring rotated beneath the feet, but the spider's eyes did not turn relative to the visual surround. Target movement and ring position were measured separately with calibrated recording devices.

This arrangement removed the retinal image motion normally produced by the animal's body rotation. A response could therefore be tested without the usual visual reduction of target error. The apparatus isolates feedback from that turn; it does not eliminate every possible sensory signal from the feet or body.



(a)

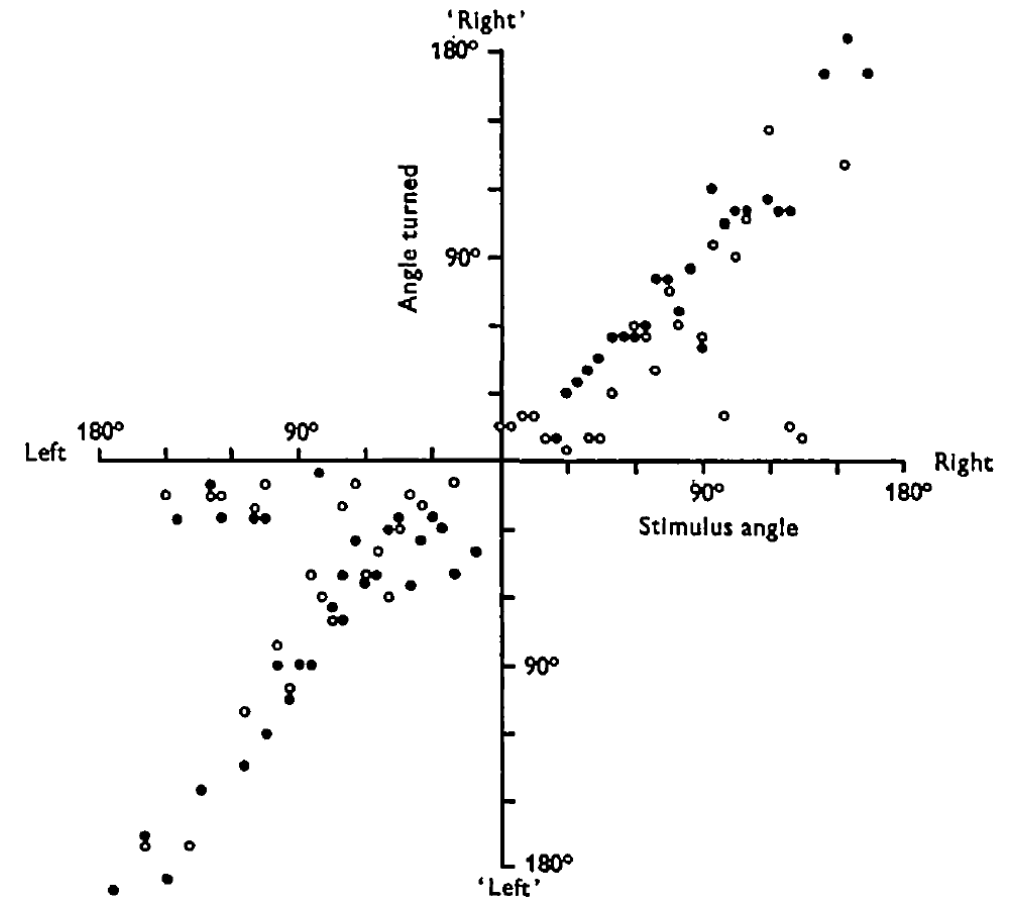
Land (1971), Fig. 2(a). Apparatus holding the body while the feet rotate a ring.

Complete turns approximate the initial target angle

An **open-loop movement** proceeds without correction from the usual feedback about its ongoing effect. Land compared attempted ring turns with the target's angular position while the spider's body remained fixed.

Many responses were complete turns whose magnitude closely approximated the initial target angle. Other responses were much smaller partial turns and did not align the forward axis with the target if interpreted as ordinary body rotations. Both occurred in the same preparation.

Accurate complete turns support a motor command based on the target position available before movement. They do not prove that natural turns never use visual feedback. The coexistence of complete and partial responses also prevents every movement from being interpreted as a perfectly specified orienting command.



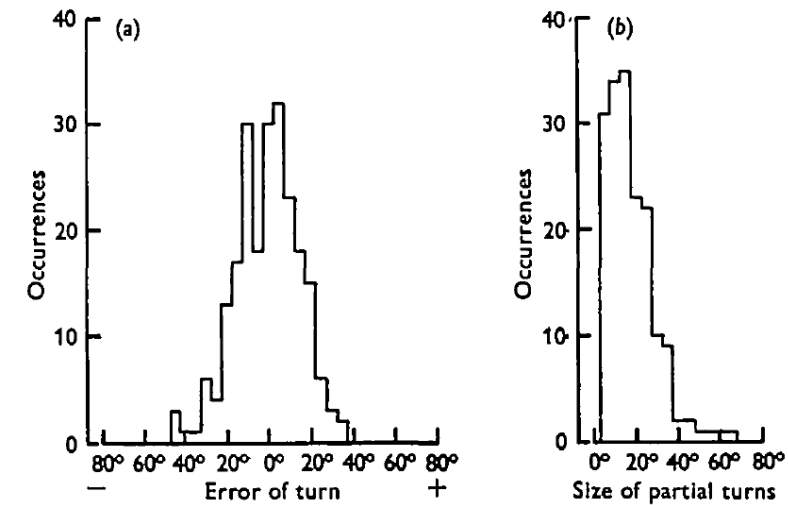
Land (1971), Fig. 5. Attempted turn magnitude versus target angle.

Turning in darkness excludes another visual correction

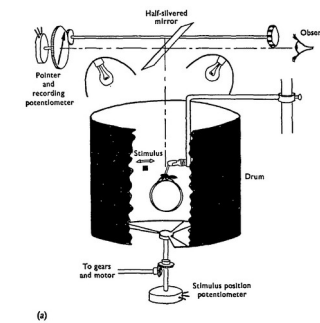
The fixed-body ring assay still allowed a spider to see the ring or its feet, potentially providing indirect visual information about its movement. Land addressed this remaining route by extinguishing illumination immediately after target motion ceased.

The stimulus normally moved 5° over 200 ms. After the light was switched off, spiders turned in darkness, and complete-turn accuracy remained comparable to the illuminated condition. Visual observation of the ring or feet was therefore not necessary for those accurate responses.

The control strengthens an open-loop account of the tested turns. It does not remove nonvisual feedback or identify the neural representation of target angle. Complete and partial movements remained behaviorally distinct, and habituation could alter which response was selected.



Land (1971), Fig. 7(a, b). Complete-turn errors and partial-turn magnitudes.



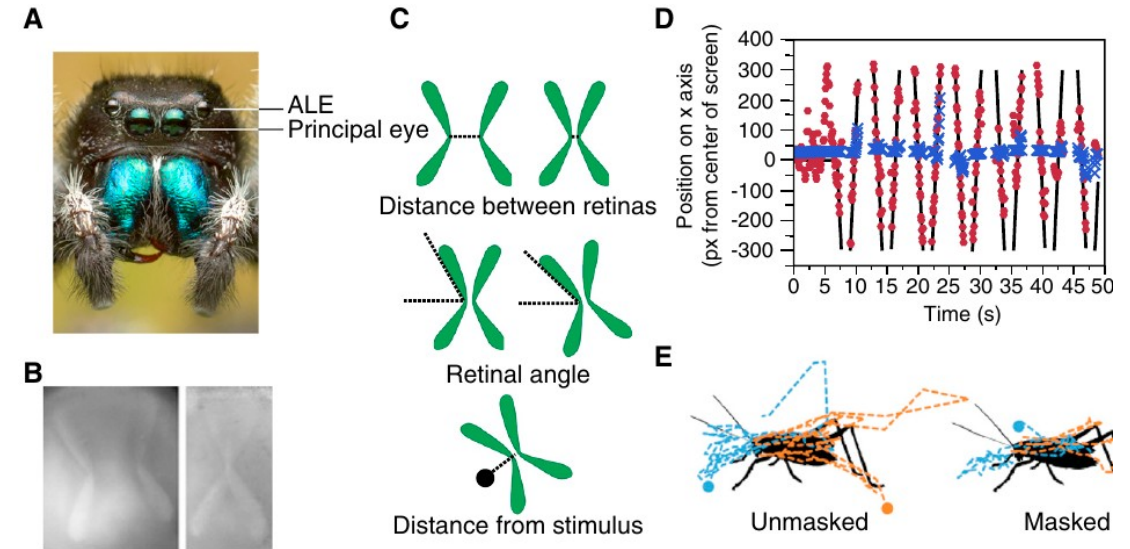
Land (1971), Fig. 2(a). The ring preparation used for feedback controls.

Lateral-eye masking disrupts principal-eye tracking

Jakob and colleagues measured the internal principal-eye movements of restrained *Phidippus audax* with an eye tracker. Moving disks were presented before and after the anterior lateral eyes were covered, testing whether the lateral pair guides the principal retina.

Disk diameters were 3° , 4° and 5° , with speeds of 2.77, 7.60 and $15.13^\circ/\text{s}$. Principal retinas followed moving targets when the lateral eyes were available. Covering that pair disrupted the target-linked movements under the same general stimulus conditions.

The manipulation supports a causal contribution of anterior lateral input to principal-eye tracking. It does not identify the connecting neurons or prove that all principal-eye movements require lateral vision. The measured motor output is internal retinal motion, rather than the body saccades in the *Servaea* ball assays.



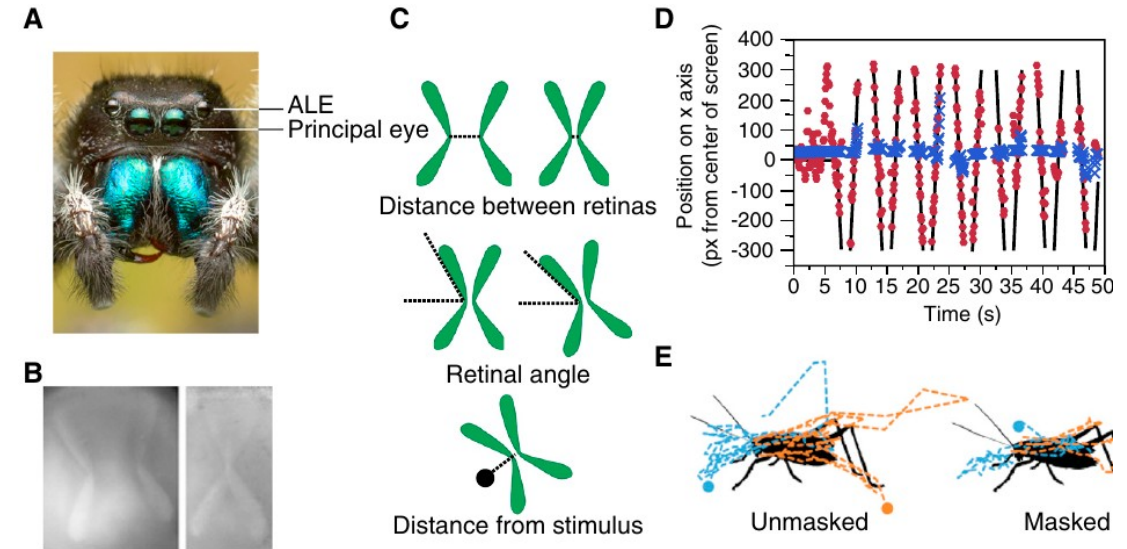
Jakob et al. (2018), Fig. 1(A–E). Principal-eye movements with anterior lateral eyes available or masked.

Masking leaves stationary-target examination available

A movement deficit can arise from sensory loss or from damage to the motor apparatus. Jakob and colleagues compared moving-target tracking with examination of stationary cricket and oval stimuli after anterior lateral masking in *Phidippus audax*.

Principal-eye scanning persisted for stationary targets even when the moving-disk tracking relationship was disrupted. The retina could still move, so the mask did not simply abolish every principal-eye motor action. Cricket and oval stimuli also recruited different degrees of target examination.

Preserved scanning narrows the interpretation toward loss of a guiding visual contribution rather than complete motor failure. It does not exclude subtler effects of masking, and the comparison does not establish which principal-eye feature signals distinguish a cricket from an oval.



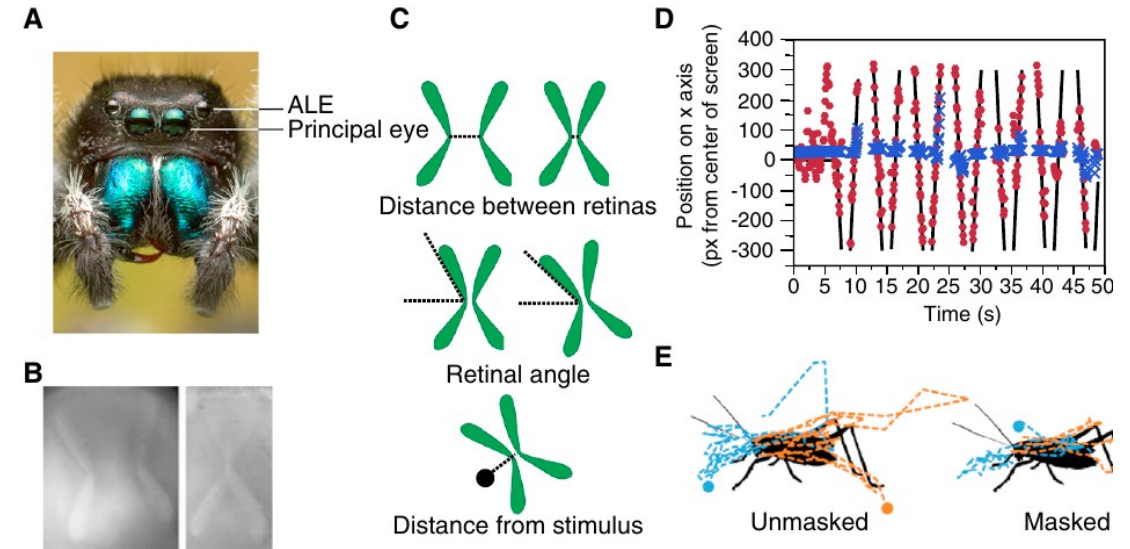
Jakob et al. (2018), Fig. 1(A–E). Moving-disk tracking and stationary-target retinal trajectories.

Retinal position and torsion are separate readouts

Jakob and colleagues quantified more than one aspect of principal-eye movement. Retinal separation and position describe where the sampled regions move, while retinal angle describes torsion, the rotation of the sampling array about the viewing axis.

The moving-target position traces were reported in the eye tracker's pixel coordinates. Stationary cricket stimuli elicited more examination and torsional movement than oval stimuli when lateral vision was available. These measures concern internal retinal behavior in restrained *Phidippus audax*.

Pixel displacement must not be relabeled as an angle without calibration. Distinct positional and torsional readouts also prevent every increase in eye movement from being called improved tracking. The data establish stimulus-dependent motor patterns while leaving the principal-eye feature representation and the underlying control signals unresolved.



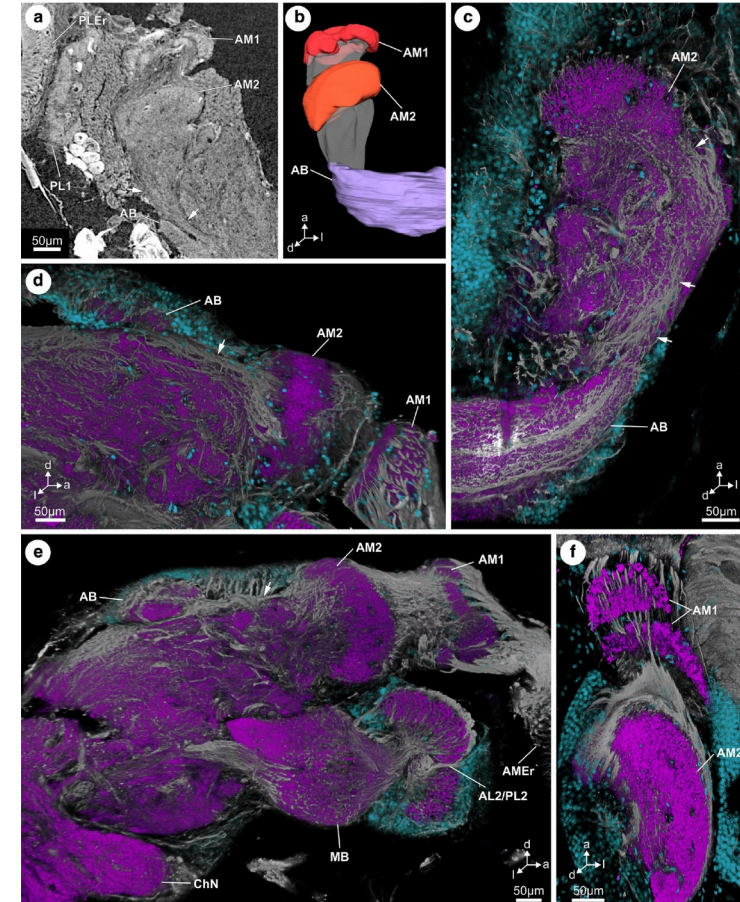
Jakob et al. (2018), Fig. 1(A–E). Distance, angle and stimulus-relative measures of retinal movement.

Principal-eye routes connect to the arcuate body

The **arcuate body** is an unpaired brain neuropil associated with visual pathways. Steinhoff and colleagues mapped the brain of *Marpissa muscosa* using tissue sections, three-dimensional imaging and molecular labeling of neuronal processes.

Principal-eye input reaches AM1, the first-order principal-eye neuropil. A thick tract connects AM1 with AM2, and a thinner tract links AM2 to the arcuate body. Antibodies against **synapsin**, a presynaptic protein, marked neuropils; **tubulin**, a structural protein, marked neuronal processes.

This anatomy supplies candidate routes from retinal signals toward central visual processing. Presynaptic-protein labeling and visible connecting processes do not establish every synapse, response sign or computation. The preparation maps *Marpissa* anatomy rather than recording the functional identity of the cells sampled in another species.



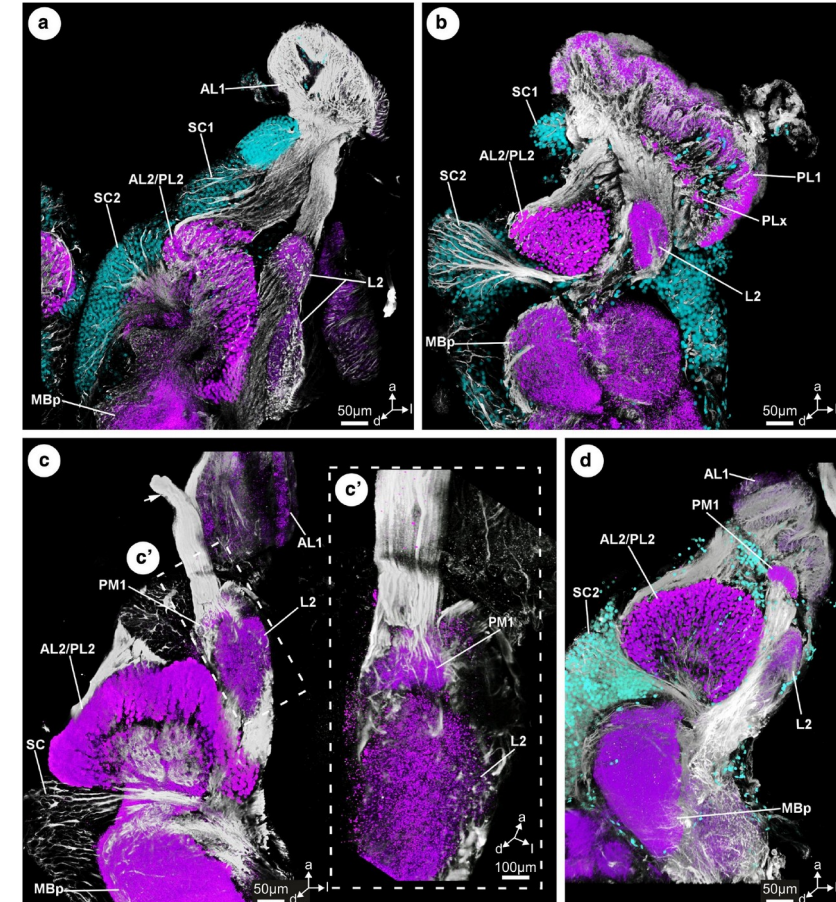
Steinhoff et al. (2020), Fig. 1(a–f). AM1, AM2 and connecting processes to the arcuate body.

Secondary-eye inputs reach shared and distinct neuropils

The **posterior lateral eyes** provide another lateral visual input, while the smaller **posterior median eyes** form a distinct pair. In *Marpissa muscosa*, Steinhoff and colleagues traced these secondary-eye routes rather than assuming they all share one motion pathway.

Anterior lateral and posterior lateral inputs reach separate first-order neuropils, AL1 and PL1, then a shared second-order region, AL2/PL2. Connections continue toward the **mushroom body**, a paired brain structure, through its stalk-like peduncle. Additional processes reach the neighboring region L2.

Posterior median input instead reaches PM1 and a tract apparently ending near the arcuate body, without an identified second-order visual neuropil. **Convergence** is the arrival of multiple inputs at a processing region. This architecture permits both separation and convergence. It does not establish the exact integration computed in L2, the mushroom body or the arcuate body.



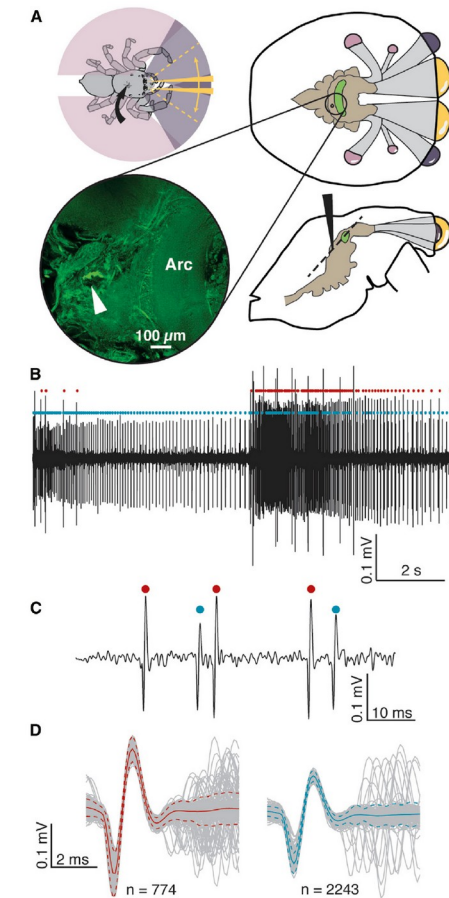
Steinhoff et al. (2020), Fig. 6(a–d). Anterior lateral, posterior lateral and posterior median routes.

Small brain openings permit extracellular recordings

An **extracellular recording** detects electrical signals near neurons without placing the electrode inside a cell. Menda and colleagues developed stable brain recordings in *Phidippus audax* by advancing a glass-insulated tungsten electrode through a small cuticular opening.

Openings of approximately 100–200 μm reduced the fluid loss that complicates access to a pressurized spider body. Visual stimulation could be paired with recordings lasting hours. Electrode positions were in or near the arcuate body, with anatomical labeling used to locate the recording region.

A **spike**, or action potential, is a brief neuronal electrical event. **Spike sorting** separates candidate units by waveform features in the extracellular signal. It supports analysis of individual response patterns but does not identify a cell's anatomy or transmitter. The restrained preparation records central visual activity without measuring the complete neural sequence during an unrestricted predatory jump.



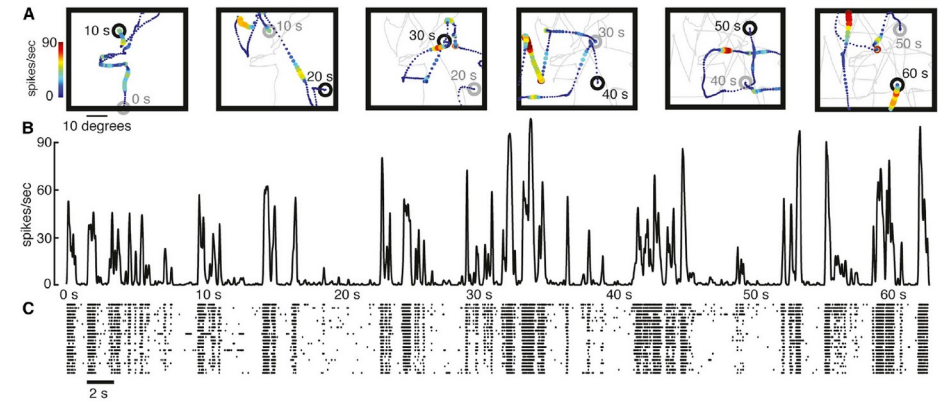
Menda et al. (2014), Fig. 1(A–D). Recording approach, arcuate-body anatomy and spike separation.

Central units respond to particular moving trajectories

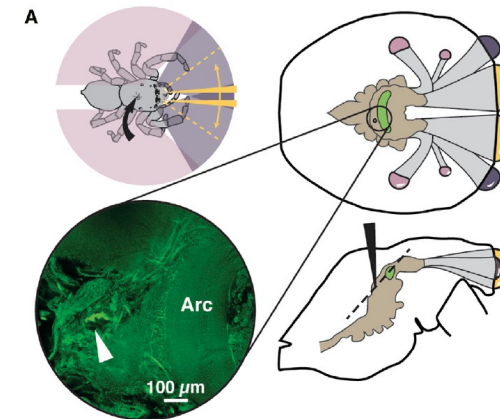
Menda and colleagues presented a small prey-like moving target while recording central units in *Phidippus audax*. A **spike raster** records action-potential times across repeated presentations, allowing temporal response consistency to be assessed.

The 1.5° target followed controlled trajectories at a display update rate of 29 frames/s. Repeated presentations evoked activity associated with particular parts of the sequence. Target position, velocity and direction were known independently of the recorded spikes.

Simple descriptions based on position or motion alone did not fully predict every response. The results support complex visual selectivity in central units, but not a uniquely defined prey-recognition circuit. A repeated response to one trajectory does not establish invariance across all target shapes, speeds or spatial positions.



Menda et al. (2014), Fig. 2(A–C). Repeated target trajectories, firing-rate traces and spike rasters.



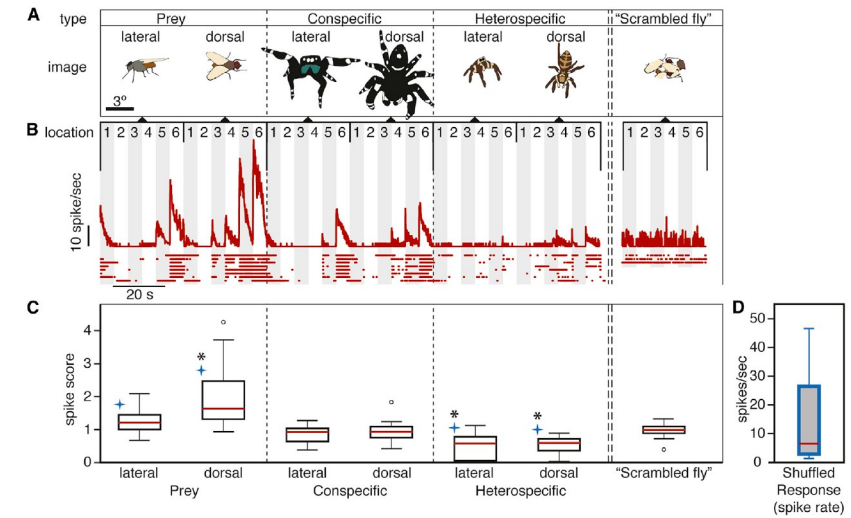
Menda et al. (2014), Fig. 1(A). The brain region approached by the recording electrode.

An intact fly image exceeds a scrambled stimulus

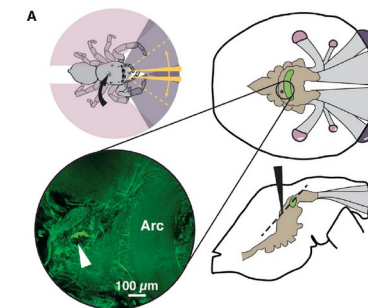
Menda and colleagues compared fly, spider and scrambled-fly images at controlled visual positions. **Scrambling** rearranged image parts while preserving elements of the original stimulus, testing whether responses depended on their organization rather than simply the presence of dark features.

In the published example unit, a dorsal fly image evoked a median 14.9 spikes/s, a lateral fly image 10.0 spikes/s and the scrambled fly 5.9 spikes/s. Responses also varied with position; images were tested at horizontal offsets of 5°, 15° and 25° on either side.

Greater activity for an intact prey configuration supports sensitivity to spatial organization. These values describe a particular recorded unit, not every spider neuron. The assay does not establish conscious recognition, identify an innate fly detector or prove that the same response directly commands an attack.



Menda et al. (2014), Fig. 3(A–D). Natural and scrambled stimuli with central-unit responses.



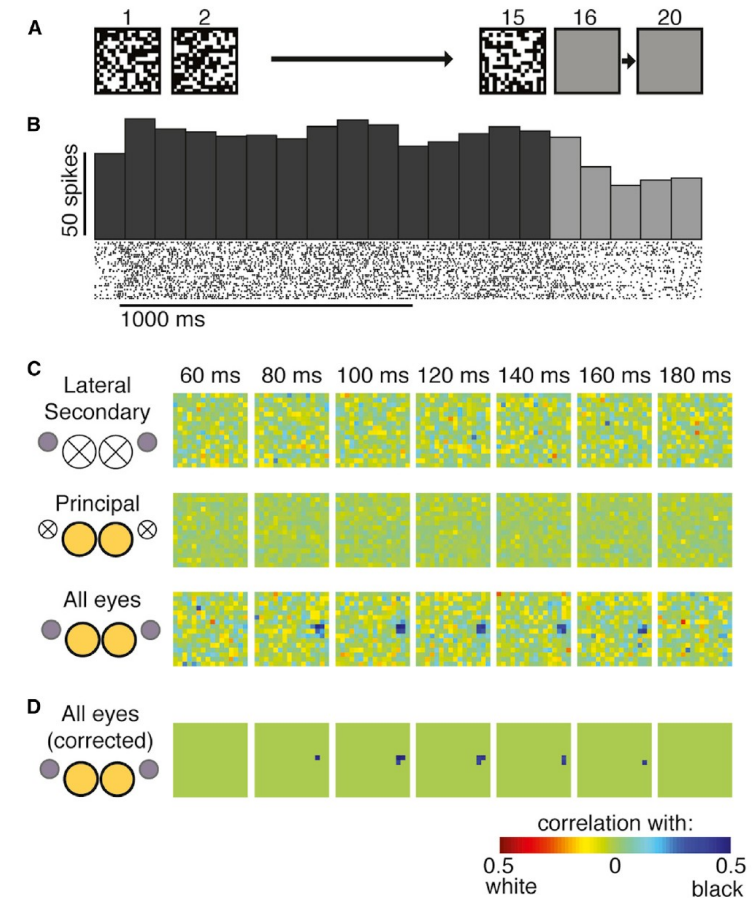
Menda et al. (2014), Fig. 1(A). Arcuate-body anatomy near the recording site.

One receptive field requires both eye systems

A **spatiotemporal receptive field** describes how stimulation at different positions and times relates to a neuron's response. Menda and colleagues estimated this relationship using rapidly changing checker patterns while selectively covering principal or lateral eyes in *Phidippus audax*.

For one recorded unit, an organized receptive field emerged with both systems available but was not recovered with either system alone. The response-related pattern occupied a temporal interval approximately 80–160 ms before the measured spikes. Internal-pattern controls reduced a display-related correlation.

The dependence supports nonlinear convergence of principal and secondary visual inputs at or before this unit. It does not identify the synapses, receptor types or exact number of processing stages. The functional result in *Phidippus* complements, but cannot be directly equated with, the anatomical pathways mapped in *Marpissa*.



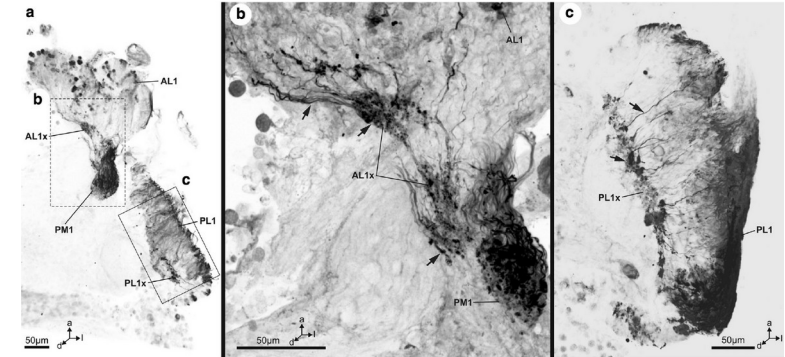
Menda et al. (2014), Fig. 4(A–D). Eye occlusion changes the measured spatiotemporal receptive field.

Histamine labels sensory axons and terminals

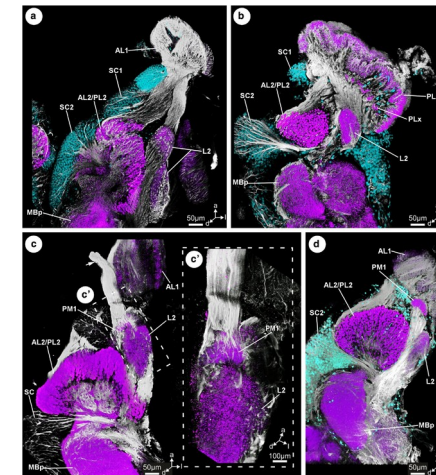
A **neurotransmitter** is a chemical released by a neuron to influence another cell. Steinhoff and colleagues used histamine antibodies to examine sensory axons and their terminal regions in *Marpissa muscosa*, linking a candidate transmitter to mapped visual anatomy.

Histamine immunoreactivity was prominent in secondary-eye axons and first-order terminal regions, including AL1, PL1 and PM1. **Immunoreactivity** means that the tissue binds an antibody recognizing the target; it is evidence of molecular presence at those locations.

The labeling supports a histaminergic photoreceptor pathway but does not measure release during light stimulation. It also does not establish the postsynaptic histamine receptor, its ion selectivity or whether the resulting current excites or inhibits a particular target. Those cellular steps must remain unresolved in this mapped spider circuit.



Steinhoff et al. (2020), Fig. 8(a–c). Histamine immunoreactivity in secondary-eye axons and terminals.



Steinhoff et al. (2020), Fig. 6(a–d). First-order neuropils receiving those sensory inputs.

Key takeaways

Complementary eye systems Lateral eyes support detection and orientation; movable principal retinas provide restricted, fine-scale visual sampling.

Layered principal retina Four receptor depths separate optical sampling; pigment identity, focus and receptor spacing are distinct properties.

Defocus-based depth *Hasarius* can jump accurately with one principal eye; red-light undershooting and intensity controls support a distance cue from image blur.

Spectral specialization Ultraviolet and green channels occur in principal eyes; *Habronattus* filtering predicts an additional red-shifted channel without a third opsin.

Active visual sampling Body saccades, retinal tracking and scanning are distinct motor outputs; lateral input guides principal-eye tracking.

Central integration and limits Mapped pathways and eye-dependent receptive fields support visual convergence, while the comparing cells, postsynaptic receptors and ion channels remain unresolved.